
How Much Can LLMs Hallucinate? An Upper Bound on Goal-Coupling Displacement

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Abstract

1 Hallucination theory bounds *frequency* from below — calibrated LLMs must hal-
2 lucinate at some rate. We bound from above a complementary quantity: the goal-
3 coupling-induced displacement of the post-update belief state — the operational
4 shape of *sycophancy*^[1], where LLM responses match user beliefs over truthful ones.
5 The chain rule of relative entropy on the post-update law equals the *transferred*
6 goal-information that enters the model state, delivering

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \sqrt{I(G; M_{\tau+} \mid e_{\tau}, M_{\tau-})}.$$

7 Čencov’s 1982 uniqueness theorem under categorical parameterization-invariance
8 pins the Fisher-Rao geometry as the canonical metric assignment, with two universal
9 constants on the resulting bound: $C = 2$ globally on the ambient spherical-arc pseu-
10 dometric — the operating constant for adversarial / rare-high-KL prompts including
11 jailbreaks and persona injection (via Rényi-1/2 monotonicity + chord-arc on the unit
12 L^2 -sphere; sharp via a symmetric N -point witness) — tightening to $C = \sqrt{2}$ locally
13 under (R) + (K) + uniform-locality on goal-conditional slices (first-moment sharp
14 via a symmetric two-point witness). A no-go theorem makes the parameterization-
15 invariance commitment load-bearing. A parallel transport-inequality track recovers
16 the canonical Stuart-school cascade form $\propto L_{\text{post}}^2 / \rho_{\text{noise}}$ across the Strand 2 hypothe-
17 sis space (tight on conjugate-Gaussian instances). Operationally the bound factors
18 as coupling strength $\times$ residual ambiguity, with architectures partitioning into a Sep-
19 arated / Partial / Coupled ladder; transformer attention and modern autoregressive
20 sequence models (linear attention, Mamba, RWKV, RetNet, long-convolutions)
21 are structurally Coupled. Binary-uniform two-goal probing gives a closed-form
22 Jensen-Shannon-divergence estimator of the transferred information. The result is
23 a conditional theory of goal-induced displacement on belief-state representations
24 under named regularity hypotheses, not a semantic metric on false outputs.

1 Introduction

26 Recent hallucination theory bounds the *frequency* with which calibrated language models must
27 produce false outputs as a consequence of training-set structure or calibration constraints —
28 Kalai and Vempala 2024^[2] on singleton-fact rates, Kalai et al. 2026^[3] on training-and-evaluation-
29 incentive origins, Karbasi et al. 2025^[4] on detection impossibility, Wu et al. 2025^[5] on VC-
30 dimension obstructions, Suzuki et al. 2025^[6] on probabilistic negligibility. Hallucination in this
31 lineage is treated as architecture-agnostic.

32 Bayesian inverse-problems theory in the lineage of Stuart 2010^[7] has spent fifteen years build-
33 ing Lipschitz-stability bounds on the posterior under perturbations of the prior, the likeli-
34 hood, and the data — Sprungk 2020^[8], Cvetković and Lie 2025^[9], Dolera and Mainini 2023^[10],

Garbuno-Iñigo et al. 2023^[11], Hosseini et al. 2025^[12]. The transport-inequality apparatus driving these bounds — Otto-Villani, Bakry-Émery, Gozlan, Bobkov-Götze — is mature and verified across decades of inverse-problems practice.

These lineages leave open the implicit belief-goal-coupling structure of attention-based architectures — a structure intrinsic to LLMs by construction (Lemma E.7), not an incidental property of training.

Engaging this structure directly reframes the question: how much can an LLM be pulled away from what its evidence supports by the goal it is asked to satisfy? The empirical referent is *sycophancy* — LLM responses matching user beliefs over truthful ones, demonstrated across state-of-the-art assistants^[1] — read as a goal-coupling-induced displacement of the response distribution from its evidence-marginal baseline; this paper bounds the magnitude of that displacement.

“Hallucination size,” in this paper, means the goal-coupling-induced displacement of the model’s evidence-conditioned belief state — not the semantic distance from a false output to truth. The two are linked through the in-context-learning correspondence (with the scope and caveats noted at Appendix E.6 and Section 6) but are not identical objects. The frequency lineage and the size axis composed here are *complementary*: a coupled-architecture model in (PI)-compatible operating regime can simultaneously have an inevitable-frequency hallucination floor *and* a bounded-magnitude goal-coupling displacement on each event.

This paper composes the two Apply the chain rule of relative entropy directly to the *post-update* law, marginalized over the goal: $\mathbb{E}_G [\text{KL}(P_{M_{\tau+}|e, M_{\tau-}, G} \| P_{M_{\tau+}|e, M_{\tau-}})] = I(G; M_{\tau+}|e_{\tau}, M_{\tau-})$. The right-hand side is the *transferred* goal-information that actually enters the post-update model state. Combined with a Fisher-Rao second-order expansion on the post-update statistical manifold, this delivers an upper bound on goal-conditional displacement:

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \cdot \sqrt{I(G; M_{\tau+}|e_{\tau}, M_{\tau-})}. \quad (1)$$

Under a parameterization-invariance commitment at full Markov-morphism strength, Čencov’s 1982 uniqueness theorem forces Fisher-Rao geometry with constant $C = 2$ globally on the ambient spherical-arc pseudometric — the operating constant for adversarial / rare-high-KL prompts (sharp via a symmetric N -point witness) — tightening to $C = \sqrt{2}$ locally under additional Riemannian structure + KL-coordinate matching + uniform-locality on goal-conditional slices. *Universal, dimension-free, no domain-specific parameters*. A scale-family no-go theorem rules out any coordinate-independent universal constant for Euclidean chart norms — making the (PI) commitment load-bearing, not coincidental. We call this *Track 2*.

A parallel *Track 1* derivation via a Talagrand transport inequality on the same post-update law generalizes the canonical Stuart-school cascade. Under Stuart-school Lipschitz-posterior hypotheses on the deterministic update map plus standard concentration on the goal-induced parameter measure and slice-wise sub-Gaussianity of the conditional data law, a Lipschitz pushforward + tensorization of T_2 (Appendix E.2) gives a CT_2 of the canonical Stuart-school form $\propto L_{\text{post}}^2 / \rho_{\text{noise}}$ on the post-update model law, with $C = \sqrt{CT_2}$ on W_2 — tight at $2L_{\text{post}}^2 / \rho_{\text{noise}}$ on conjugate-Gaussian and exponential-family-conjugate instances, with an absolute multiplicative overhead from the kernel-tensorization step in general. The Class 1 (Separated) specialization of Track 1’s hypothesis space is the Stuart-school setup; Track 1 extends the cascade to the goal-coupled regime outside Stuart-school’s framing. Track 2’s universal constant is what makes the composition novel; Track 1 connects directly to the Bayesian-inverse-problems lineage at the cascade level. Both flow from the same chain-rule move on the post-update law.

The architectural reading Under a coupling-attenuation hypothesis (H_{κ}), the bound factors:

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \cdot \sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega_{\tau} | e_{\tau}, M_{\tau-})}, \quad (2)$$

into an *architectural coupling factor* $\kappa_{\text{processing}}$ (how much of the available goal-information the architecture’s update mechanism transfers) and a *residual ambiguity factor* $I(G; \Omega | e, M)$ (how much the goal could in principle resolve about the latent world-state beyond what the evidence already pins down). Architectures partition by their goal-update topology into a monotonic ladder — Class

1 (Separated), Class 2 (Partial), Class 3 (Coupled). For Class 1, $\kappa_{\text{processing}} \leq 1$ holds automatically by data-processing on the Markov chain $G \rightarrow \Omega \rightarrow M_{\tau+}$. For Class 2 and Class 3, (H_{κ}) is a named structural commitment with documented sufficient conditions and a worst-case witness (parrot architecture, $\kappa_{\text{processing}} = \infty$) excluded by scope.

Plain decoder-only transformer attention is *Class 3 (Coupled) by construction* in a precise structural sense: every output computation downstream of the goal positions has a directed-graph path from at least one goal position whenever attention weights are non-degenerate on the goal row (Lemma E.7, by induction on layer depth, robust to RMSNorm / FlashAttention / causal masking / sliding-window). The structural classification is at the *architectural* level — what attention *can* do; the *quantitative* coupling magnitude $\kappa_{\text{processing}}$ requires additional assumptions and is empirically estimated, not analytically derived. The structural Coupled-class claim does not depend on the token-distribution-as-belief-state idealization that the bias-bound *application* additionally invokes via the in-context-learning correspondence^[13–16].

Contributions

1. **An architectural classification.** Goal/Update Coupling Class — Class 1 (Separated), Class 2 (Partial), Class 3 (Coupled) — partitions architectures monotonically by goal-update topology, grounded in the Pearl-blanket reading^[17] of the Markov-blanket apparatus. Lemma E.7 derives Class 3 status for transformer attention from connectivity alone.
2. **An umbrella upper bound** (1) on goal-conditional displacement of the post-update model state, derived via chain rule on the post-update law. Two named tracks supply C — a transport-inequality cascade (Track 1) and a Fisher-Rao geometry route (Track 2).
3. **A no-go that forces a coordinate commitment.** No coordinate-independent universal constant exists for Euclidean chart norms (a scale-family construction). Under (PI) alone, Rényi-1/2 monotonicity + the chord-arc identity give a sharp universal $C = 2$ globally on the ambient spherical-arc pseudometric; under the additional (R) + (K) triple plus uniform-locality, Čencov’s uniqueness theorem tightens this to Fisher-Rao + $\sqrt{2}$ locally — both forced, neither free.
4. **An architectural-corollary form** (2) under (H_{κ}) , recovering the $\kappa \times \mathcal{A}$ factorization that gives the bound its operational reading.

Section 3 sets up the architectures and the (PI), (R), (K) axioms. Section 4 states the umbrella theorem with both tracks, the no-go, and the architectural corollary. Section 5 sketches the mechanism — both proofs flow from the post-update chain rule. Limitations and the bound’s relation to the frequency-lower-bound lineage are in Section 6.

2 Related Work

Two literatures sit on either side of the gap this paper bridges; brief categorization (full positioning across all six adjacent strands in Appendix F).

LLM hallucination theory has produced architecturally-agnostic frequency lower bounds on calibrated language models — calibration-constrained singleton-fact rates, training-and-evaluation-incentive origins, learnability and computability obstructions, probabilistic-negligibility positive results (cited in §1). These results derive from training-set structure or capacity considerations and do not engage the architecture’s belief-goal-coupling structure. They bound *how often* a calibrated model must hallucinate.

Bayesian inverse-problems posterior stability (the Stuart school) has produced mature Lipschitz-continuity bounds on the posterior under perturbations of the prior, the likelihood, and the data, using transport-inequality machinery (Otto-Villani, Bakry-Émery, Gozlan, Bobkov-Götze) and Fisher-information-flavored constants in optimal-transport frameworks (cited in §1). These results address well-posedness of Bayesian inference; they have not been applied to belief-goal-coupled inference. The machinery bounds *how stable* a Bayesian posterior is to specified perturbations.

The two bound different things, sit in different research traditions, and do not engage with each other. The chain-rule of relative entropy applied directly to the post-update law (Lemma 5.1) is the bridging move: it converts the goal-coupling problem into a transferred-information quantity, which

the Stuart-school transport machinery and the Fisher-Rao expansion both bound at the post-update law. *Both* lineages enter — Track 1 inherits the Otto-Villani / Bakry-Émery cascade; Track 2 deploys Čencov uniqueness on the manifold the Stuart school operates on. The composition is the contribution; the architectural reading via Class 1 / Class 2 / Class 3 makes the bound operational.

Adjacent strands — Fisher-Rao geometry for Bayesian sensitivity analysis, Owhadi-Scovel-Sullivan brittleness no-gos, mean-field variational inference stability, the Markov-blanket apparatus from active inference — sit close enough to require explicit cite-and-distinguish but do not resolve the gap. Detailed positioning across all strands is in Appendix F.

3 Setup

We work with goal-conditioned probabilistic models in a Bayesian-update setting. The state at time τ^- before update consists of an epistemic component M_{τ^-} (the model state, an element of a parameter manifold $\mathcal{M}$) and a goal component G . An event e_τ arrives — a token, an observation, a query response — and the model updates to M_{τ^+} . The latent world-state Ω_τ is what the event is evidence about. The architecture’s update mechanism produces a post-update model state M_{τ^+} whose law depends — possibly stochastically — on the goal G . The goal-conditional post-update law is $P_{M_{\tau^+}|e_\tau, M_{\tau^-}, G}$; the goal-marginal post-update law $P_{M_{\tau^+}|e_\tau, M_{\tau^-}}$ is the natural baseline against which goal-coupling is measured (the operating-distribution average over goals).

3.1 The bias quantity

The *goal-conditional bias* measures the displacement of the goal-conditional law from the goal-marginal:

$$\|\Delta M_{\text{bias}}(G)\| := d_{\mathcal{M}}\left(P_{M_{\tau^+}|e_\tau, M_{\tau^-}, G}, P_{M_{\tau^+}|e_\tau, M_{\tau^-}}\right), \quad (3)$$

a random variable in G . Throughout, $d_{\mathcal{M}}(\cdot, \cdot)$ denotes one of two distinct objects, used cleanly within their respective tracks. (*Track 1*) the Wasserstein distance W_2 on **laws over the parameter manifold** $\mathcal{M}$ — the goal-conditional and goal-marginal post-update random elements of $\mathcal{M}$ are compared as probability laws on $\mathcal{M}$ with a ground metric on $\mathcal{M}$. (*Track 2*) the Fisher-Rao spherical-arc distance d_{FR} on the statistical-manifold sub-case (H1), where each model state $M_t \in \mathcal{M}$ carries an associated probability distribution P_{M_t} over the latent world-state — and the bias quantity is the random spherical arc between $P_{M_{\tau^+}|e_\tau, M_{\tau^-}, G}$ and $P_{M_{\tau^+}|e_\tau, M_{\tau^-}}$ as points on the unit L^2 -sphere of $\sqrt{p}$. We adopt the *Amari-Nagaoka spherical-arc convention* $d_{FR}(P, Q) := 2 \arccos \int \sqrt{pq}$ throughout, with $d_{FR} \in [0, \pi]$. (Equal to the intrinsic Fisher-Rao geodesic on the full simplex / nonparametric ambient space; on a parametric submanifold the ambient spherical arc is the chart-invariant pseudometric we bound.) The locally-tight $\sqrt{2}$ regime (under (PI)+(R)+(K)+(H4')) operates in the small-displacement limit where the ambient arc agrees with submanifold-intrinsic geodesics at second order; the globally-valid $C = 2$ regime (under (PI) alone) operates on the ambient arc directly. Euclidean-on-parameters is the choice Section 4.1 rules out.

3.2 Goal/Update Coupling Class

Architectures partition by whether G has a causal path into the belief-update computation. The partition is monotonic in coupling strength.

Class 1 (Separated) is the architectural-separation case studied throughout Bayesian inverse-problems theory. Class 3 (Coupled) is the case the existing hallucination-theory lineage doesn’t engage geometrically. Class 2 (Partial) is genuinely intermediate — hybrid LLM systems with retrieval, tools, scratchpads, or external memory — and lives at the system level rather than the architecture level. The Pearl-blanket reading^[17] of the Markov-blanket apparatus grounds the partition; we do not adopt the Friston-blanket metaphysical reading.

The bound’s architectural-corollary form (2) factors transferred goal-information through an *attenuation ratio* — the proportion of in-principle-resolvable goal-information that the architecture’s update mechanism actually transfers into the post-update model state:

Table 1: Goal/Update Coupling Class — partition by conditional-independence structure.

Class	Goal/update coupling	Examples
Class 1 (Separated)	Holds by construction — estimator has no causal path from G	Kalman filter + LQR; modular RL with separate world model
Class 2 (Partial)	Holds for some pathways, fails for others	Biological cortex; hybrid systems with separate preprocessing
Class 3 (Coupled)	Fails by construction — G is causally upstream of every downstream output computation	Autoregressive sequence models with causally-aggregated cross-position mixer (transformers, linear attention, Mamba/SSMs, RWKV, RetNet, long-convolutions — Lemma E.7 + Corollary E.8)

$$\kappa_{\text{processing}} := \frac{I(G; M_{\tau+} \mid e_{\tau}, M_{\tau-})}{I(G; \Omega_{\tau} \mid e_{\tau}, M_{\tau-})}. \quad (4)$$

179 Conditioning on $M_{\tau-}$ matters: prior correlation between goals and prior model state — present even
 180 in Class 1 — would inflate the measure otherwise. *Class 1 formally:* conditional on $(e_{\tau}, M_{\tau-})$, $G \perp\!\!\!\perp$
 181 $M_{\tau+} \mid \Omega_{\tau}$ — G affects $M_{\tau+}$ only through the latent world-state Ω_{τ} . This conditional independence
 182 is the formal content of “estimator has no causal path from G ” and gives the Markov chain $G \rightarrow$
 183 $\Omega_{\tau} \rightarrow M_{\tau+}$ (conditional on $(e_{\tau}, M_{\tau-})$), to which the data-processing inequality applies, yielding
 184 $\kappa_{\text{processing}} \leq 1$ automatically. For Class 2 and Class 3, $\kappa_{\text{processing}} \leq 1$ is a named structural commitment;
 185 the parrot architecture $M_{\tau+} := G$ with $G \perp\!\!\!\perp \Omega_{\tau}$ given $(e_{\tau}, M_{\tau-})$ produces $\kappa_{\text{processing}} = \infty$, explicitly
 186 outside scope. *Convention.* When both numerator and denominator vanish (no transferred goal-
 187 information *and* no residual ambiguity, e.g., a Class 1 architecture on goal-orthogonal evidence), set
 188 $\kappa_{\text{processing}} := 0$; the bound’s right-hand side is then trivially zero.

189 3.3 Axioms

190 The universal-constant route (Section 5.2, Track 2) commits to three axioms on the bound’s metric
 191 $d_{\mathcal{M}}$:

192 **Hypothesis 3.1** ((PI) Sufficient-statistic invariance across the natural category). *The bias bound’s*
 193 *distance / divergence is a natural assignment on the category of standard parametric statistical*
 194 *models — for each model, a candidate distance or divergence — commuting with congruent Markov*
 195 *morphisms (sufficient statistics) in the sense of Čencov 1982^[18]. The induced object on $\mathcal{M}$ inherits*
 196 *this categorical assignment.*

197 (PI) at full Markov-morphism strength is the *categorical* invariance Čencov’s uniqueness theorem
 198 requires: the assignment must respect every sufficient-statistic morphism across the category, not
 199 merely smooth coordinate changes on a single fixed $\mathcal{M}$. (PI) alone admits both Riemannian metrics
 200 and f -divergences as candidates (the global Fisher-Rao spherical-arc bound and the Hellinger backstop
 201 in Appendix D are the two (PI)-only theorems); (R) below adds the Riemannian-metric specifier
 202 needed for Čencov’s local uniqueness statement.

203 **Hypothesis 3.2** ((R) Riemannian structure). *The bound’s metric $d_{\mathcal{M}}$ derives from a smooth Riemannian*
 204 *metric on $\mathcal{M}$.*

205 (R) restricts the (PI)-invariant candidates to Riemannian metrics specifically, filtering out divergence-
 206 form alternatives (TV, χ^2 , Hellinger-as-divergence) that are (PI)-invariant as f -divergences but are
 207 not metric distances in the Riemannian sense.

208 **Hypothesis 3.3** ((K) KL second-order matching). $\text{KL}(P_{\theta} \parallel P_{\theta+\delta}) = \frac{1}{2} d_{\mathcal{M}}^2(P_{\theta}, P_{\theta+\delta}) + O(d_{\mathcal{M}}^3)$.

209 (K) fixes the global scalar Čencov leaves free, pinning the bound’s constant C at $\sqrt{2}$. (R) and (K)
 210 are both implicit in any “ $\mathbb{E} d \leq C\sqrt{T}$ ” theorem-shape with second-order matching; we name them
 211 explicitly so the uniqueness statement of Theorem 4.5 has its hypotheses fully stated.

The transport-inequality route (Section 5.1, Track 1) and the dimension-restricted slice-wise refinements (Theorem 4.1’s Track 2 locally-tight instantiation) introduce additional regularity hypotheses inline with the theorems that invoke them. Throughout the paper we restrict to the *statistical-manifold sub-case*:

Hypothesis 3.4 ((H1) Statistical-manifold sub-case). *The model state $M_t \in \mathcal{M}$ corresponds to a probability distribution P_{M_t} over latent world-states, and $\mathcal{M}$ is locally a statistical manifold in the sense of Amari and Nagaoka 2000^[19]; $\mathcal{M}$, the goal-space, and the event-space are standard Borel, so regular conditional probabilities exist as Markov kernels^[20] Theorem 6.3.*

(H1) is satisfied by parametric Bayesian posterior families, exponential families, conjugate priors, and the parametric-belief-state subclass of LLMs that admits a natural distributional reading. The standard-Borel regularity is what makes the post-update chain rule (used in both Tracks) hold in the abstract-spaces form^[21] Theorem 3.4; ^[22] Theorem 5.4.

Coupled-class autoregressive connectivity (deferred). Lemma E.7 establishes that plain decoder-only transformer attention — and the broader autoregressive-sequence-model family with causally-aggregated cross-position mixers under a per-source non-degeneracy condition (linear attention, Mamba/SSMs, RWKV, RetNet, long-convolutions; named-architecture verifications + ICL-correspondence caveat in Appendix E.6) — is structurally Class 3 by directed-graph reachability. The claim is on $\kappa_{\text{processing}}$ ’s *support*, not its magnitude.

4 Main Results

The umbrella theorem bounds the goal-conditional displacement of the post-update model state by transferred goal-information. Two named tracks supply the constant — a transport-inequality cascade (Track 1) and a Fisher-Rao geometry route (Track 2) — both flowing from a common chain-rule move on the post-update law (mechanism in Section 5).

Theorem 4.1 (Umbrella bound on goal-conditional displacement). *Under (H1) and the regularity hypotheses below, for fixed event-prior pair $(e_\tau, M_{\tau-})$ with $\mathbb{E}$ taken over $G \sim P(G | e_\tau, M_{\tau-})$:*

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \cdot \sqrt{I(G; M_{\tau+} | e_\tau, M_{\tau-})}, \quad (5)$$

with C supplied by either of two routes:

(Track 1, transport-inequality): *Under additional (H2') — a Talagrand T_2 inequality on the post-update law — $C = \sqrt{C_{T_2}}$ with metric W_2 . The constant C_{T_2} recovers the canonical Stuart-school cascade value $2L_{\text{post}}^2 / \rho_{\text{LSI}}$ as a special case under standard log-Sobolev concentration.*

(Track 2, Fisher-Rao spherical-arc): *The bound holds with metric d_{FR} in the ambient Amari-Nagaoka spherical-arc convention of Section 3.1 ($d_{FR}(P, Q) = 2 \arccos \int \sqrt{pq}$, the great-circle arc on the unit L^2 -sphere of $\sqrt{p}$). Two universal forms — both dimension-free, neither carrying domain-specific parameters:*

locally tight — *under (PI)+(R)+(K) on the natural categorical metric assignment plus (H4') at parameter δ_* , $C = \sqrt{2(1 + R(\delta_*))}$ with explicit $R(\delta_*) \rightarrow 0$ as $\delta_* \rightarrow 0$ (locally tight at $\sqrt{2}$ in the limit; Čencov uniqueness pins the metric and constant via Theorem 4.5);*

globally valid — *under (PI) alone (no (R), no (K), no (H4')), the ambient spherical-arc bound $C = 2$ holds throughout — sharp via a symmetric N -point witness as $N \rightarrow \infty$ (Theorem D.1 in Appendix D; the $\sqrt{2}$ overhead is exactly the chord-arc factor at the unit L^2 -sphere’s antipode in squared-distance form). On parametric submanifolds the ambient spherical arc is the chart-invariant pseudometric; submanifold-intrinsic geodesics may be longer (the local-tight regime is exactly where ambient and intrinsic agree at second order).*

Hypothesis 4.2 ((H2') Talagrand T_2 on the post-update law). $W_2^2(P, P_{M_{\tau+}|e, M_{\tau-}}) \leq C_{T_2} \text{KL}(P \| P_{M_{\tau+}|e, M_{\tau-}})$ for all P . *Sufficient conditions in Appendix B (LSI, Otto-Villani; dimension-free sub-Gaussian Lipschitz concentration^[23]).*

Hypothesis 4.3 ((H4') Uniform local regime). $\text{ess sup}_g \text{KL}(P_{M_{\tau+}|G=g} \| P_{M_{\tau+}}) \leq \delta_*$ for some $\delta_* \in (0, 1)$ — *every goal-conditional slice uniformly close to the goal-marginal, not merely the*

average (strictly stronger than small- I_M). Adversarial / rare-high-KL prompts (jailbreaks, persona injection) exit ($H4'$); the global $C = 2$ bound (Theorem D.1) is the operating tool there.

261 **Remarks on Theorem 4.1.**

262 *The two tracks bound different metrics with the same square-root-in-information shape.* Track 1 lives on
263 W_2 , Track 2 on d_{FR} . Both deliver bounds of form $\mathbb{E} d \leq C\sqrt{I_M}$ where $I_M = I(G; M_{\tau+} | e_{\tau}, M_{\tau-})$
264 is transferred goal-information. The two are not “different scalings”; they are bounds on different
265 metrics with the same information dependence.

266 *The bound is unconditional in the architectural classification.* Both Tracks bound transferred goal-
267 information directly. (H1)–(H2') and (H1)–(H4')–(PI)–(R)–(K) are regularity hypotheses on the
268 post-update law and the metric. Neither makes a structural commitment about how the goal G enters
269 the architecture’s update mechanism. The architectural reading enters at corollary level via (H_{κ})
270 below.

271 *Track 2’s constant is universal and dimension-free.* Track 1’s C_{T_2} is family-specific (depends on
272 the post-update concentration). Track 2’s $\sqrt{2}$ is the same constant across the entire categorical
273 metric assignment under the (PI)+(R)+(K) triple, by Čencov uniqueness (Theorem 4.5 below); on the
274 parametric submanifold of any specific application, the induced metric is Fisher-Rao with the same
275 constant.

276 *Track 1 generalizes the Stuart-school cascade.* Under Stuart-school Lipschitz-posterior hypothe-
277 ses^[7,8,10,12] plus T_2 concentration on the goal-induced parameter measure and slice-wise conditional
278 sub-Gaussianity, Theorem E.3 yields the canonical $C_{T_2} \propto L_{\text{post}}^2 / \rho_{\text{noise}}$ — tight at $2L_{\text{post}}^2 / \rho_{\text{noise}}$ on
279 conjugate-Gaussian and exponential-family-conjugate instances (Appendix C); Track 1 extends the
280 cascade beyond Class 1 (Separated) to goal-coupled architectures, a regime Stuart-school does not
281 address.

282 **4.1 A no-go that forces (PI)**

283 The universal-constant route (Track 2’s $\sqrt{2}$) is not coincidental. A scale-family construction shows
284 that no coordinate-independent universal constant exists for Euclidean chart norms — making the
285 (PI) commitment load-bearing.

286 **Theorem 4.4** (No-go on Euclidean chart norms). *Suppose there were a constant $C_0 < \infty$ —*
287 *independent of any chart on any statistical manifold — such that for every chart ϕ on $\mathcal{M}$ and*
288 *every goal-coupled architecture satisfying (H1)+(H2') with non-zero post-update displacement,*
289 $\mathbb{E} \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi} \leq C_0 \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}$. *Then no such C_0 exists.*

290 The proof is short: a chart rescaling $\phi \mapsto a\phi$ scales the chart-Euclidean Wasserstein distance linearly
291 while leaving the right-hand side (chart-invariant information) unchanged; taking $a \rightarrow \infty$ contradicts
292 the fixed $C_0\sqrt{I}$. Full statement and proof in Appendix E.4. The Gaussian scale family with chart σ
293 versus $\log \sigma$ is the cleanest illustration.

294 **Remark** Theorem 4.4 is *adjacent in shape but opposite in direction* to Owhadi-Scovel-Sullivan’s
295 Bayesian-brittleness results^[24,25]. Their no-go is for *arbitrary stability under finite information* in
296 fixed metrics; ours is for *a universal constant absent a coordinate-invariance commitment*, in a setting
297 where (H2')’s post-update concentration explicitly excludes their brittleness regime. The two no-gos
298 constrain different things; neither implies the other.

299 **4.2 Čencov uniqueness and sharpness**

300 The no-go requires a coordinate-invariance commitment if a universal constant is wanted at all;
301 (PI)+(R)+(K) is the natural commitment for an information-coordinate cascade; and Čencov unique-
302 ness within that commitment leaves little further freedom.

303 **Theorem 4.5** (Uniqueness and sharpness of the Fisher-Rao $+\sqrt{2}$ bound (local)). *Under*
304 *(H1)+(PI)+(R)+(K)+(H4'), in the umbrella form (5) (where $\|\Delta M_{\text{bias}}\|$ is the $d_{\mathcal{M}}$ -displacement*
305 *per (3)): (a) $d_{\mathcal{M}}$ is uniquely the ambient Amari-Nagaoka Fisher-Rao spherical-arc distance, agreeing*
306 *with submanifold-intrinsic Fisher-Rao geodesics at second order in the ($H4'$) regime; (b) $C = \sqrt{2}$ is*
307 *the unique sharp upper-bound constant — no $C < \sqrt{2}$ holds uniformly across the ($H4'$) regime.*

Part (a) follows from Čencov’s 1982 uniqueness theorem applied to the natural categorical metric assignment of (PI). Part (b) is verified by a symmetric two-point witness on the conjugate-Gaussian Class 1 family: $G \in \{-a, +a\}$ uniform delivers $\mathbb{E} d_{FR}/\sqrt{I_M} \rightarrow \sqrt{2}$ as $a/\sigma \rightarrow 0$ (the symmetry forces d_{FR} exactly constant across slices, so Jensen is tight). Full proof + witness construction in Appendix E.5. Alternative chart-independent commitments (TV, Hellinger-as-divergence, χ^2 , Rényi) are all defeated by (R)/(K) or non-tightness; the no-go narrows the candidate space, and (PI)+(R)+(K) selects Fisher-Rao + $\sqrt{2}$.

4.3 The architectural corollary

The umbrella theorem bounds displacement by *transferred* goal-information. The architectural reading factors this transferred information into a structural-architectural quantity (where can the goal flow?) and an information-theoretic quantity (how much could the goal in principle resolve?).

Hypothesis 4.6 ((H_κ) Bounded architectural attenuation). $\kappa_{\text{processing}} \leq \kappa^* < \infty$, with κ^* a named architectural constant (typically 1 for non-amplifying architectures).

Corollary 4.7 (Architectural factorization). Under (H1) + the Track-specific hypotheses + (H_κ) at level κ^* , the bound factors as

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \sqrt{\kappa^* \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau^-})}, \quad (6)$$

with C supplied by Track 1 or Track 2 as in Theorem 4.1.

Remarks on Corollary 4.7.

The factorization is the contribution-headline. Coupling strength $\kappa_{\text{processing}} \times$ residual ambiguity $I(G; \Omega \mid e, M)$ — architecture $\times$ information. Theorem 4.1’s unconditional form is the structural backbone whose math holds without (H_κ) ; the corollary is the operational reading.

Class 1 (Separated) gives (H_κ) automatically. The Markov chain $G \rightarrow \Omega_\tau \rightarrow M_{\tau+}$ (conditional on $(e_\tau, M_{\tau-})$) plus data-processing inequality yields $\kappa_{\text{processing}} \leq 1$ by construction. (H_κ) is a derived consequence, not a commitment, for Class 1.

Class 2 (Partial) and Class 3 (Coupled) require (H_κ) as a named structural commitment. Two clean sufficient conditions and a parrot-architecture worst-case witness ($\kappa_{\text{processing}} = \infty$ when $M_{\tau+} := G$ on goals orthogonal to the latent world-state) are documented in Appendix B.3. The unconditional theorem still bounds parrot-architecture displacement directly by transferred information; the $\kappa \times \mathcal{A}$ factorization just doesn’t apply.

Transferred information has a closed-form empirical estimator. For $G \in \{g_1, g_2\}$ uniform, $I(G; M_{\tau+} \mid e_\tau, M_{\tau-}) = \text{JSD}(P_{M_{\tau+}|g_1}, P_{M_{\tau+}|g_2})$ — Monte-Carlo estimable from per-goal response samples; the local Track-2 form gives $\mathbb{E} \|\Delta M_{\text{bias}}\| \leq \sqrt{2 \text{JSD}}$, the global form $\mathbb{E} d_{FR} \leq 2\sqrt{\text{JSD}}$ (Appendix D). Factorization additionally requires lower-bounding $I(G; \Omega \mid e, M_{\tau-})$ — architecture-side modeling, Section 6.

5 Mechanism

Both tracks bridge to the bound through the same move: a chain rule of relative entropy applied directly to the post-update law, marginalized over the goal. This is the structural composition of the two literatures — the Bayesian-inverse-problems posterior-stability machinery and the architectural classification meet at this identity.

Lemma 5.1 (Chain rule on the post-update law). Under (H1), $\mathbb{E}_G [\text{KL}(P_{M_{\tau+}|e, M_{\tau-}, G} \parallel P_{M_{\tau+}|e, M_{\tau-}})] = I(G; M_{\tau+} \mid e_\tau, M_{\tau-})$.

The identity is exact (Polyanskiy and Wu²¹, Theorem 3.4; Gray²², Theorem 5.4 for the abstract-spaces form). The right-hand side is *transferred* goal-information — the actually-realized coupling between G and the post-update model state. Combined with either a transport inequality on the post-update law or a Fisher-Rao second-order expansion of KL, the bound follows by taking expectation over G and applying Jensen.

353 5.1 Track 1 — transport-inequality cascade

354 Apply (H2') slice-wise: $W_2^2(P_{M_{\tau+}|G=g}, P_{M_{\tau+}}) \leq C_{T_2} \cdot \text{KL}_g$. Take expectation over G , substitute
 355 Lemma 5.1 for the slice-wise mean, and apply Jensen on $\sqrt{\cdot}$: $\mathbb{E} W_2 \leq \sqrt{C_{T_2} I_M}$. The cascade
 356 composes textbook information-theoretic and transport-inequality machinery — Otto-Villani for the
 357 LSI-implies- T_2 direction, Bakry-Émery for K -strong-log-concavity-implies-LSI, Gozlan’s dimension-
 358 free sub-Gaussian Lipschitz-concentration characterization. The contribution is the application to the
 359 goal-conditional bias quantity on the post-update law: there is no separate posterior-pushforward step,
 360 because the chain rule lives on the post-update law directly. Full proof + (H2') sufficient-condition
 361 verifications in Appendix B.

362 5.2 Track 2 — Fisher-Rao expansion

363 The KL-to-Fisher-Rao second-order expansion on the ambient L^2 -sphere of $\sqrt{p}$ — $\text{KL}(P \parallel Q) =$
 364 $\frac{1}{2} d_{FR}^2(P, Q) + O(d_{FR}^3)$ in the Amari-Nagaoka spherical-arc convention (Amari and Nagaoka¹⁹ §3.7)
 365 — applies to nearby points on the ambient sphere. The ambient framing absorbs the goal-marginal-
 366 baseline-as-mixture concern: both the slice law $P_{M_{\tau+}|G=g}$ and the goal-marginal mixture $P_{M_{\tau+}}$ are
 367 points on the ambient sphere by construction. Apply slice-wise under (H4'), take $\mathbb{E}_G$, substitute
 368 Lemma 5.1, and apply Jensen on $\sqrt{\cdot}$: $\mathbb{E} d_{FR} \leq \sqrt{2} \sqrt{I_M} (1 + o(1))$. The $(1 + o(1))$ remainder is
 369 controlled uniformly under (H4') by the Amari-Chentsov tensor and minimum Fisher eigenvalue (full
 370 bound in Appendix E.3). The constant $\sqrt{2}$ is what (PI)+(R)+(K) buys: Čencov uniqueness pins it
 371 (Theorem 4.5).

372 *Outside (H4')*. Track 2 has globally-valid companion bounds (ambient FR spherical-arc at $C = 2$,
 373 Hellinger chord at $1/\sqrt{2}$) under (H1)+(PI) only — full taxonomy and chord-arc-at-antipode geometric
 374 reading in Appendix D.

375 5.3 No-go — chart rescaling

376 The no-go’s proof is a chart-rescaling argument: $\phi \mapsto a\phi$ scales the chart-Euclidean W_2 linearly while
 377 leaving KL, MI, Fisher-Rao spherical-arc, and Hellinger chart-invariant; taking $a \rightarrow \infty$ contradicts
 378 any candidate fixed $C_0\sqrt{I}$. Gaussian scale charts σ vs. $\log \sigma$ illustrate. Constructively: universal
 379 constants require coordinate-invariant norms — under (PI)+(R)+(K), Čencov uniqueness selects
 380 Fisher-Rao + $\sqrt{2}$ (Appendix E.4).

381 6 Conclusion

382 The opening question — *how much can an LLM be pulled away from its evidence by its goal?* — sits
 383 on an axis the frequency lineage doesn’t engage. A coupled-architecture model can simultaneously
 384 have an inevitable-frequency hallucination floor (training-data structure, calibration, capacity) *and* a
 385 bounded-magnitude goal-coupling displacement on each event — orthogonal axes, complementary
 386 readings.

387 **Limitations** Conditional theorems under named regularity ((H2'), (H4'), (H $_{\kappa}$)) and the (PI) ax-
 388 iomatic commitment, on theory-only grounds — one channel of hallucination, complementary to the
 389 frequency lineage.

390 **Future directions** *First instantiation of $\hat{\kappa}_{processing}$ on a deployed LLM*: the bound becomes op-
 391 erational once κ^* is bounded; binary-uniform two-goal probing has a closed-form JSD estimator
 392 (Theorem 4.1), with activation-level mediation via representation-engineering / steering-vector /
 393 mechanistic-interpretability for the architectural side. *Sycophancy*^[1] — LLM responses matching
 394 user beliefs over truthful ones across state-of-the-art assistants — is the natural empirical referent;
 395 Sharma et al.’s *answer sycophancy metric* is essentially the binary-uniform two-goal probe of Corol-
 396 lary 4.7, so instantiating the bound on existing sycophancy benchmarks would convert them into
 397 magnitude-bound diagnostics. *Bridging to the frequency-lower-bound lineage*: a layered theory com-
 398 posing Kalai-Vempala-style frequency-floors with the goal-coupling magnitude-ceiling here would
 399 give the field a complete two-axis picture.

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A Failed routes

We document two derivation routes that failed at structural levels, so future work does not re-attempt them without new evidence.

A.1 Route F1: Cramér-Rao inversion

Attempt Treat G as a parameter to be estimated from Ω given (e, M) . The Cramér-Rao lower bound gives $\text{Var}(\hat{G}) \geq \mathbf{I}_G^{-1}$ where $\mathbf{I}_G$ is the Fisher information about G given (Ω, e, M) . Invert this to a posterior-sensitivity factor: an upper bound $\|\Delta M_{\text{bias}}\| \leq L_M \cdot \sqrt{\mathbf{I}_G^{-1}}$ via a Lipschitz-in- G constant on the coupled-update mechanism.

Why it fails Cramér-Rao bounds estimator variance from *below*. To produce an *upper bound* on $\|\Delta M_{\text{bias}}\|$ via Cramér-Rao inversion would require fixing an estimator class within which $\mathbf{I}_G^{-1}$ is also an upper bound on variance — but no such estimator class exists for the coupled architecture’s update mechanism. The mechanism is a deterministic function of (G, e, M) , not an estimator of G , and there is no uniform-class assumption available. The direction of the Cramér-Rao bound and the direction of the bias bound are opposite; inverting one to the other requires an additional assumption that is not available.

Lesson Information-theoretic *lower-bound* machinery for estimator performance does not invert to upper bounds on side-information-induced displacement.

A.2 Route F2: Rate-distortion inversion

Attempt Apply Shannon’s rate-distortion inequality $R(D) \geq h(X) - \frac{1}{2} \log(2\pi e D)$ for a Gaussian source. Invert to $D \leq \sigma^2 \cdot 2^{-2(R-h(X))}$ with “bits in” interpreted as $I(G; \Omega|e, M)$, “distortion” interpreted as $\|\Delta M_{\text{bias}}\|^2$.

Why it fails Rate-distortion theory describes the minimum bits-per-symbol required to *represent* a source within distortion D ; it does not describe the maximum *displacement induced* by injecting side-information into an update mechanism. The two problems have different structures: source-coding is about optimal representations; side-information injection is about perturbation propagation. Identifying “bits in” with I and “distortion” with displacement does not produce a sound inequality — the rate-distortion curve is about a different operational quantity.

Lesson Source-coding theorems are about optimal representations of a source, not spatial displacement induced by side-information injection. Upper bounds on displacement require transport-inequality machinery (Pinsker, Otto-Villani, Bakry-Émery, Lipschitz-posterior).

A.3 The pattern

Both failed routes attempt to invert *information-theoretic lower-bound* machinery (Cramér-Rao for estimator error; rate-distortion for source coding) into *upper bounds on displacement*. The structural mismatch is uniform: lower-bound machinery for optimization quantities does not invert to upper bounds on perturbation quantities without additional structure that is not available in coupled-architecture settings. Transport-inequality machinery is on the other side of this divide and is the correct family for upper-bound-on-displacement results.

B Hypothesis verification details

This appendix collects the regularity-condition verifications referenced in Section 5.1 and Section 5.2.

B.1 (H2’) Talagrand T_2 inequality on the post-update law — sufficient conditions

(H2’) requires $W_2^2(P, P_{M_{\tau+}|e, M_{\tau-}}) \leq C_{T_2} \text{KL}(P \| P_{M_{\tau+}|e, M_{\tau-}})$ for all probability measures P . Three clean sufficient conditions:

Route (a): LSI on the post-update law If $P_{M_{\tau+}|e, M_{\tau-}}$ satisfies a logarithmic Sobolev inequality with constant $\rho_{\text{post}} > 0$,

$$\text{Ent}_{P_{M_{\tau+}|e, M_{\tau-}}}(f^2) \leq \frac{2}{\rho_{\text{post}}} \mathbb{E}_{P_{M_{\tau+}|e, M_{\tau-}}} [\|\nabla f\|^2] \quad \text{for all sufficiently smooth } f,$$

then by^[26] Theorem 1, (H2’) holds with $C_{T_2} = 2/\rho_{\text{post}}$. A sufficient condition for LSI, by Bakry-Émery’s curvature-dimension criterion^[27], is K -strong log-concavity of the post-update density: $-\nabla^2 \log p_{M_{\tau+}|e, M_{\tau-}} \succeq K \cdot I$ for some $K > 0$, giving $\rho_{\text{post}} \geq K$.

Hypothesis (S) — own proof of strong log-concavity for Bayesian post-updates When the post-update law is itself a Bayesian posterior with negative log-likelihood $\Phi(\theta; \omega)$ and prior density $\pi(\theta)$, the posterior negative log-density is $U(\theta; \omega) = \Phi(\theta; \omega) - \log \pi(\theta)$. Strong log-concavity of the posterior is a *direct* statement about the Hessian-in- θ of U : $\nabla_{\theta}^2 U(\theta; \omega) \succeq K_{\text{eff}} \cdot I$ uniformly in $\omega \in \Omega$, for some $K_{\text{eff}} > 0$. Two clean sufficient conditions:

- (S-i) *Convex likelihood + strongly log-concave prior.* If $\nabla_{\theta}^2 \Phi(\theta; \omega) \succeq 0$ uniformly in (θ, ω) (likelihood log-concave in θ) and π is K -strongly log-concave (so $-\nabla_{\theta}^2 \log \pi \succeq K \cdot I$), then $\nabla_{\theta}^2 U \succeq K \cdot I$, giving $K_{\text{eff}} \geq K$.
- (S-ii) *Bounded negative likelihood curvature dominated by prior curvature.* If $\nabla_{\theta}^2 \Phi(\theta; \omega) \succeq -K_0 \cdot I$ uniformly with $K_0 < K$ (likelihood may be locally non-convex but its negative-curvature is bounded), and π is K -strongly log-concave, then $\nabla_{\theta}^2 U \succeq (K - K_0) \cdot I$, giving $K_{\text{eff}} \geq K - K_0 > 0$.

Either condition gives Bakry-Émery $\rho_{\text{post}} \geq K_{\text{eff}}$ and Otto-Villani $C_{T_2} \leq 2/K_{\text{eff}}$. Sprungk 2020^[8] derives sharper local-Lipschitz constants for the posterior across TV / Hellinger / W_2 / KL; Cvetković and Lie 2025^[9] establish matched lower bounds; Dolera and Mainini 2023^[10] give global W_2 -Lipschitz constants in terms of Fisher-information functionals — which can sharpen C_{T_2} beyond the bare $2/K_{\text{eff}}$. Mixed-Hessian smoothness $\|\partial_{\theta\omega}^2 \Phi\| \leq H$ is *not* a substitute: it controls how the posterior mode shifts with ω , but the LSI / strong-convexity claim is about the curvature in θ and must be made directly.

Route (b) (corrected — bounded support gives T_1 , not T_2). Bare bounded support of diameter D on $\mathcal{M}$ is **not** a T_2 -sufficient condition. *Counterexample.* On $\{0, 1\} \subset \mathbb{R}$ (diameter $D = 1$), let $\nu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$ and $\mu_{\epsilon} = (\frac{1}{2} - \epsilon)\delta_0 + (\frac{1}{2} + \epsilon)\delta_1$ for $\epsilon \in (0, 1/2)$. Then $W_2^2(\mu_{\epsilon}, \nu) = \epsilon$ (the optimal coupling moves mass ϵ from 0 to 1), while $\text{KL}(\mu_{\epsilon} \parallel \nu) = 2\epsilon^2 + O(\epsilon^4)$, so $W_2^2/\text{KL} = 1/(2\epsilon) + O(1) \rightarrow \infty$ as $\epsilon \rightarrow 0^+$ — no constant C uniformizes the ratio. The structural reason: Gozlan 2009^[23] established that T_2 is *equivalent* to dimension-free Gaussian concentration of $\mu^{\otimes n}$, strictly stronger than diameter (Talagrand-type product concentration requires LSI / curvature input, not just bounded support).

What does follow from diameter is the *weaker* T_1 inequality: by $W_1 \leq D \cdot \text{TV}$ and Pinsker $\text{TV} \leq \sqrt{\text{KL}/2}$,

$$W_1^2(P, P_{M_{\tau+|e}, M_{\tau-}}) \leq C_{T_1} \text{KL}(P \parallel P_{M_{\tau+|e}, M_{\tau-}}), \quad C_{T_1} = D^2/2$$

(squared convention matching (H2'); equivalently $W_1 \leq (D/\sqrt{2})\sqrt{\text{KL}}$, sharp on the two-point Bernoulli at $p = 1/2$ — Bobkov and Götze²⁸ with Hoeffding sub-Gaussian variance proxy $\sigma^2 = D^2/4$ for distributions on a diameter- D set, applied to the T_1 -equivalent Lipschitz-sub-Gaussian characterization). Substituting W_1 for W_2 in Step 2 of the Section 5.1 cascade preserves the *distance-form* bound $\mathbb{E}\|\Delta M_{\text{bias}}\| \leq (D/\sqrt{2})\sqrt{I(G; M_{\tau+|e}, M_{\tau-})}$ at the T_1 level — at the cost of giving up Theorem 4.1's tighter squared-distance / linear-in- I refinement (5). Token-distribution belief states for LLMs on the simplex are more naturally treated under Track 2 (Fisher-Rao, Section 5.2) than via Track 1's diameter-only T_1 weakening; in finite-dimensional bounded-away-from-boundary simplex regimes, the universal $\sqrt{2}\sqrt{I}$ form holds.

Route (c): dimension-free sub-Gaussian Lipschitz concentration Per Gozlan 2009^[23], μ satisfies $T_2(C)$ iff $\mu^{\otimes n}$ enjoys dimension-free Gaussian concentration of Lipschitz functionals: there exist constants $C', \lambda > 0$ such that $\mathbb{E}_{\mu^{\otimes n}}[\exp(\lambda(f - \mathbb{E}f)^2)] \leq 2$ uniformly in n for every 1-Lipschitz $f: \mathbb{R}^n \rightarrow \mathbb{R}$. Bobkov and Götze 1999^[28] characterizes the strictly weaker *single-Lipschitz-function* version (giving T_1); the Gozlan strengthening to T_2 is what makes single-distribution sub-Gaussian moments insufficient on their own. When dimension-free product concentration holds, C_{T_2} admits an explicit form in the variance proxy.

When both routes fail Heavy-tailed post-updates without LSI and without dimension-free sub-Gaussian concentration are outside Track 1's T_2 scope. The T_1 fallback under bounded support (Route (b) as corrected above) recovers the distance form at weaker constants — adequate for $\mathbb{E}\|\Delta M_{\text{bias}}\| \leq C\sqrt{I}$. Track 2 under (PI) + (R) + (K) does not require (H2') and remains available; Theorem D.2's Hellinger backstop is a global universal-constant fallback under (PI) only.

672 B.2 KL-to-Fisher second-order expansion

673 For nearby distributions $P_\theta, P_{\theta+\delta}$ on a parametric family with Fisher information $\mathbf{I}(\theta)$:

$$\text{KL}(P_{\theta+\delta} \parallel P_\theta) = \frac{1}{2} \delta^\top \mathbf{I}(\theta) \delta + O(\|\delta\|^3) = \frac{1}{2} d_{FR}^2(P_\theta, P_{\theta+\delta}) + O(d_{FR}^3).$$

674 Standard derivation: Cover and Thomas²⁹ §12.5; Amari and Nagaoka¹⁹ §3.7 Theorem 3.1. The
675 $O(d_{FR}^3)$ remainder is sharp in the sense that the second-order term is exact and the third-order term
676 is generically non-zero (off-exponential-family).

677 B.3 $(\mathbf{H}_\kappa)$ sufficient conditions

678 The architectural-corollary commitment $(\mathbf{H}_\kappa)$ is that $\kappa_{\text{processing}} =$
679 $I(G; M_{\tau+} | e, M_{\tau-}) / I(G; \Omega_\tau | e, M_{\tau-})$ is bounded by some named architectural constant $\kappa^* < \infty$
680 (typically 1). For Class 1 (Separated) architectures, the data-processing inequality applied to the
681 Markov chain $G \rightarrow \Omega_\tau \rightarrow M_{\tau+}$ (conditional on $(e_\tau, M_{\tau-})$ by architectural separation) gives
682 $\kappa_{\text{processing}} \leq 1$ automatically. For Class 2 (Partial) and Class 3 (Coupled), two structural conditions
683 each give $\kappa_{\text{processing}} \leq 1$ (or $\leq 1 + \alpha$ in the second condition):

684 1. *Goal-blind effective kernel.* Suppose the post-update map factors as $M_{\tau+} =$
685 $K(M_{\tau-}, e_\tau, \tilde{\Omega}_\tau)$ with K deterministic, where $\tilde{\Omega}_\tau$ is a goal-conditional reweighting of
686 Ω_τ that is *non-amplifying*: $I(G; \tilde{\Omega}_\tau | e_\tau, M_{\tau-}) \leq I(G; \Omega_\tau | e_\tau, M_{\tau-})$. Then conditional
687 on $(e_\tau, M_{\tau-})$, $G \rightarrow \tilde{\Omega}_\tau \rightarrow M_{\tau+}$ is a Markov chain (since K depends on G only through
688 $\tilde{\Omega}_\tau$), and DPI applied along this chain gives

$$I(G; M_{\tau+} | e_\tau, M_{\tau-}) \leq I(G; \tilde{\Omega}_\tau | e_\tau, M_{\tau-}) \leq I(G; \Omega_\tau | e_\tau, M_{\tau-}),$$

689 i.e., $\kappa_{\text{processing}} \leq 1$.

690 1. *Bounded direct architectural channel.* Suppose the architecture’s *direct* goal-coupling
691 pathway (independent of Ω_τ) has bounded conditional MI relative to the resolvable chan-
692 nel: $I(G; M_{\tau+} | e_\tau, M_{\tau-}, \Omega_\tau) \leq \alpha I(G; \Omega_\tau | e_\tau, M_{\tau-})$ for some $\alpha \geq 0$. The chain
693 rule of mutual information conditioned on $(e_\tau, M_{\tau-})$ gives $I(G; M_{\tau+}, \Omega_\tau | e_\tau, M_{\tau-}) =$
694 $I(G; \Omega_\tau | e_\tau, M_{\tau-}) + I(G; M_{\tau+} | e_\tau, M_{\tau-}, \Omega_\tau)$, and using $I(G; M_{\tau+} | e_\tau, M_{\tau-}) \leq$
695 $I(G; M_{\tau+}, \Omega_\tau | e_\tau, M_{\tau-})$ (additional side information cannot decrease MI), we get
696 $\kappa_{\text{processing}} \leq 1 + \alpha$. The case $\alpha = 0$ — architectural channel adds nothing beyond what Ω_τ
697 provides given $(e_\tau, M_{\tau-})$ — gives $\kappa_{\text{processing}} \leq 1$.

698 **Worst-case witness — parrot architecture** Consider a degenerate Coupled-class architecture
699 whose update rule is $M_{\tau+} := G$ (the post-update model state is identically the goal). Two computa-
700 tions:

701 *Numerator.* Since $M_{\tau+} = G$ deterministically conditional on $(e_\tau, M_{\tau-})$, $I(G; M_{\tau+} | e_\tau, M_{\tau-}) =$
702 $H(G | e_\tau, M_{\tau-})$ exactly.

703 *Denominator.* Suppose $G \perp\!\!\!\perp \Omega_\tau$ given $(e_\tau, M_{\tau-})$ — the goal carries no information about the latent
704 world-state beyond what the evidence and prior have already pinned down. (Operationally this is the
705 “make me feel good” goal of Section 3.2: the user’s affective goal is independent of the empirical
706 question the evidence resolves.) Then $I(G; \Omega_\tau | e_\tau, M_{\tau-}) = 0$.

707 Hence $\kappa_{\text{processing}} = H(G | e_\tau, M_{\tau-}) / 0 = +\infty$. $(\mathbf{H}_\kappa)$ explicitly fails. This shows the architectural
708 commitment is *real* for non-Separated architectures, not a free move. The unconditional theorems
709 (Theorem 4.1, Theorem D.1, Theorem D.2) still bound the parrot architecture’s displacement by
710 transferred information directly — the prefactor is $\sqrt{H(G | e_\tau, M_{\tau-})}$ rather than $\sqrt{\kappa I(G; \Omega)}$ — but
711 the $\kappa \times \mathcal{A}$ factorization (and hence the architectural-corollary form) does not apply.

C Conjugate-Gaussian numerical comparison

We compute the Track-family constants explicitly for the conjugate-Gaussian observation-prior pair and exhibit the regimes in which they are tight. *The setup separates the agent's internal model from the operating distribution that probes goal-coupling.* The agent uses the conjugate Gaussian prior/likelihood update rule with prior $\theta \sim \mathcal{N}(0, \tau^2)$ and observation channel $\Omega|\theta \sim \mathcal{N}(\theta, \sigma^2)$. The external operating distribution used to probe goal-coupling has $\Omega|G = g \sim \mathcal{N}(\beta(g), \sigma^2)$ — the goal directly shifts the operating-data mean. The architecture's update is goal-blind (Class 1 Separated, conjugate Bayesian update). Identifying $M_{\tau+}$ with the posterior-mean parameter $\mu_+ := L_{\text{post}}\Omega$ (variance $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ fixed), the post-update parameter law under operating G is $\mu_+|G \sim \mathcal{N}(L_{\text{post}}\beta(G), L_{\text{post}}^2\sigma^2)$.

Two distinct variances. The *inside-posterior covariance* $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ is the spread of the posterior measure on θ at a fixed observation. The *parameter-law variance* $L_{\text{post}}^2\sigma^2 = L_{\text{post}}\sigma_{\text{post}}^2$ is the spread of the random model state μ_+ under data noise. They differ by exactly L_{post} . Track 1's C_{T_2} is determined by the parameter-law variance, not the inside-posterior covariance; conflating them is the algebraic error this appendix corrects.

Transferred information and $\kappa_{\text{processing}}$. The goal-conditional law of Ω has small- $\text{Var}_G(\beta)$ MI $I(G; \Omega|e, M_{\tau-}) \approx \text{Var}_G(\beta)/(2\sigma^2)$. The post-update parameter law inherits the same MI: the deterministic linear map $\Omega \mapsto L_{\text{post}}\Omega$ is invertible, so the data-processing inequality on the chain $G \rightarrow \Omega \rightarrow M_{\tau+}$ saturates and $I(G; M_{\tau+}|e, M_{\tau-}) = I(G; \Omega|e, M_{\tau-})$ exactly. The realized attenuation ratio is $\kappa_{\text{processing}}^{\text{realized}} = 1$. *This Class 1 conjugate-Gaussian setup does not exhibit architectural attenuation* — it sits at the worst-case Class 1 corner where DPI is tight; non-trivial $\kappa < 1$ requires a non-invertible data channel (e.g., quantization, dimension reduction) or a Class 2/3 architecture.

Track 1 (Theorem 4.1). Route (a) — LSI on the post-update parameter law $\mathcal{N}(0, L_{\text{post}}^2\sigma^2)$ at small $\text{Var}_G(\beta)$ — gives $\rho_{\text{post}} = 1/(L_{\text{post}}^2\sigma^2)$ and

$$C_{T_2} = 2/\rho_{\text{post}} = 2L_{\text{post}}^2\sigma^2 = \frac{2\tau^4\sigma^2}{(\sigma^2 + \tau^2)^2}.$$

Theorem 4.1: $\mathbb{E}[W_2^2] \leq C_{T_2} \cdot I(G; M_{\tau+}|e, M_{\tau-}) = 2L_{\text{post}}^2\sigma^2 \cdot I(G; \Omega|e, M_{\tau-})$ (using $\kappa = 1$). Equivalently, identifying $\rho_{\text{noise}} := 1/\sigma^2$ as the *noise concentration* of the conditional data law $\Omega | \theta$ (the quantity that enters the Stuart-school cascade's denominator — distinct from the LSI constant of the marginal observation law $P_\Omega \sim \mathcal{N}(0, \sigma^2 + \tau^2)$, which would be $1/(\sigma^2 + \tau^2)$),

$$\mathbb{E}[W_2^2] \leq \frac{2L_{\text{post}}^2}{\rho_{\text{noise}}} \cdot I(G; \Omega|e, M_{\tau-})$$

— the canonical Stuart-school cascade constant, recovered as C_{T_2} alone (DPI saturation $\kappa = 1$ is the witness that this Class 1 example sits at the Stuart corner). The constant $2L_{\text{post}}^2\sigma^2$ here is read off *directly* from the post-update parameter law's variance $\mathcal{N}(0, L_{\text{post}}^2\sigma^2)$ — sub-Gaussian by inspection. The general structural reduction Theorem E.3 across the broader Stuart-school hypothesis space recovers a constant of the same canonical form $\propto L_{\text{post}}^2/\rho_{\text{noise}}$, with an absolute multiplicative overhead from the kernel-tensorization step; the conjugate-Gaussian instance lands at the *tight* constant via the direct read-off rather than via the abstract reduction. The cascade is sharp here: the slice-wise W_2^2/KL ratio equals C_{T_2} exactly for these Gaussian parameter laws. Limit checks: prior-dominant ($\tau \ll \sigma$): $C_{T_2} \approx 2\tau^4/\sigma^2 \rightarrow 0$; likelihood-dominant ($\sigma \ll \tau$): $C_{T_2} \approx 2\sigma^2 \rightarrow 0$; balanced ($\tau = \sigma = 1$): $C_{T_2} = 1/2$ and $\mathbb{E}[W_2^2] \leq I_M/2$.

Track 2 (Theorem 4.1). Fisher metric on the parameter-law manifold $\{\mathcal{N}(\mu, L_{\text{post}}^2\sigma^2) : \mu \in \mathbb{R}\}$ (laws over the random parameter μ_+ , not posterior measures on θ): $\mathbf{I}(\mu) = 1/(L_{\text{post}}^2\sigma^2)$. Fisher-Rao distance $d_{FR}(\mathcal{N}(\mu_1, v), \mathcal{N}(\mu_2, v)) = |\mu_1 - \mu_2|/\sqrt{v} = |\mu_1 - \mu_2|/(L_{\text{post}}\sigma)$. Slice-wise to the marginal $\mathcal{N}(0, L_{\text{post}}^2\sigma^2)$: $d_{FR}(P_{M_{\tau+}|G=g}, P_{M_{\tau+}}) \approx |\beta(g)|/\sigma$ at small $\text{Var}_G(\beta)$. Theorem 4.1 (unconditional): $\mathbb{E}[d_{FR}^2] \leq 2 \cdot I(G; M_{\tau+}|e, M_{\tau-}) = \text{Var}_G(\beta)/\sigma^2$ — sharp on this family.

Track 2 Euclidean translation (Theorem D.4). $|\Delta\mu_+|_{\text{Eucl}} = L_{\text{post}}\sigma \cdot d_{FR}$, so $\mathbb{E}|\Delta\mu_+|_{\text{Eucl}} \leq L_{\text{post}}\sigma\sqrt{2I_M}$. The prefactor $L_{\text{post}}\sigma = \tau^2\sigma/(\sigma^2 + \tau^2)$ is bounded above by $\tau/2$ uniformly (AM-GM

at $\sigma = \tau$) and *vanishes in both extreme limits*: $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow \infty$ (very noisy observations: the architecture learns essentially nothing from Ω , posterior-mean displacement vanishes); $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow 0$ (sharp observations: $L_{\text{post}} \rightarrow 1$ but $\sigma \rightarrow 0$). Maximum at balanced $\sigma = \tau$. The naive “ $\sigma\sqrt{2I}$ ” reading is wrong (correct prefactor is $L_{\text{post}}\sigma$, not bare σ); equally importantly, the “ $\sigma_{\text{post}}\sqrt{2I}$ ” reading is *also* wrong (it conflates inside-posterior covariance $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ with parameter-law variance $L_{\text{post}}^2\sigma^2$).

Hellinger backstop (Theorem D.2). Theorem D.2: $\mathbb{E} \text{Hel}(P_{M_{\tau+}|G}, P_{M_{\tau+}}) \leq (1/\sqrt{2})\sqrt{I_M}$ globally, where Hel is between two parameter laws $\mathcal{N}(\mu_1, L_{\text{post}}^2\sigma^2)$ and $\mathcal{N}(\mu_2, L_{\text{post}}^2\sigma^2)$. Hellinger is globally bounded by 1, so the bound is non-trivial when $\sqrt{I_M}/2 < 1$ — i.e., $I_M < 2$ nat.

Numerical comparison at balanced scale ($\tau = \sigma = 1$, $L_{\text{post}} = 1/2$, $L_{\text{post}}\sigma = 1/2$, $L_{\text{post}}^2\sigma^2 = 1/4$, transferred info $I_M = 1$ nat for normalization):

Table 2: Conjugate-Gaussian numerical comparison at balanced scale ($\tau = \sigma = 1$, $I_M = 1$ nat).

Track	Theorem	Bound	Numerical at $I_M = 1$
Track 1	Theorem 4.1	$\mathbb{E}[W_2^2] \leq C_{T_2} I_M = (1/2) I_M$	$\mathbb{E} W_2 \leq 1/\sqrt{2}$
Track 2	Theorem 4.1	$\mathbb{E} d_{FR} \leq \sqrt{2I_M}$	$\mathbb{E} d_{FR} \leq \sqrt{2}$
Track 2 Euclidean translation	Theorem D.4	$\mathbb{E} \ \Delta\mu_+\ \leq L_{\text{post}}\sigma\sqrt{2I_M} = (1/2)\sqrt{2I_M}$	$\mathbb{E} \ \Delta\mu_+\ \leq 1/\sqrt{2}$
Hellinger	Theorem D.2	$\mathbb{E} \text{Hel} \leq \sqrt{I_M}/2$	$\mathbb{E} \text{Hel} \leq 1/\sqrt{2}$

Track 1 and the Track 2 Euclidean translation agree exactly at balanced scale ($\mathbb{E} W_2 = \mathbb{E} \|\Delta\mu_+\|_{\text{Eucl}} = 1/\sqrt{2}$ at $I_M = 1$) — they bound the same Euclidean displacement; the Track 2 Euclidean translation is Track 2 with the Fisher-Rao-to-Euclidean conversion factor $L_{\text{post}}\sigma$ on the parameter chart. The Euclidean translation goes to zero in both extreme limits (vanishing prior-dominant and vanishing likelihood-dominant). At $I_M = 1$ nat the Hellinger bound saturates the global $\sqrt{1/2} \approx 0.71$; for $I_M \ll 1$ Track 1 / Track 2 Euclidean are tighter; for $I_M \gtrsim 2$ the Hellinger backstop is the only non-vacuous bound.

D Track 2 companions and parametric Euclidean translations

The locally-tight Fisher-Rao geodesic bound (Theorem 4.1’s Track 2 instantiation under (PI)+(R)+(K)+(H4’)) is one of three Fisher-Rao-spine bounds; the other two — a globally-valid *ambient Fisher-Rao spherical-arc* bound and a Hellinger chord backstop — recover under strictly weaker hypotheses, with universal constants of their own. Together with the conjugate-Gaussian Euclidean translation (clarifying the no-go’s apparent scale-family pathology) and the exponential-family Euclidean generalization, they form the Track 2 family.

D.1 Track 2 global Fisher-Rao spherical-arc backstop

Outside the uniform-locality regime — when a few rare goals carry disproportionately large transferred information, or when the operating distribution sits at moderate-to-large I_M — the slice-wise expansion’s $(1 + o(1))$ remainder is no longer uniformly sharp. A global *ambient spherical-arc* bound is available under strictly weaker hypotheses, at the cost of a slightly larger constant. (The bound is on the Bhattacharyya / Fisher spherical-arc pseudometric $d_{FR} = 2 \arccos(\text{BC})$ on the unit L^2 -sphere of $\sqrt{p}$, which equals intrinsic Fisher-Rao only on the ambient simplex / full nonparametric distribution space; on parametric submanifolds it is the chart-invariant pseudometric, $\leq$ submanifold-intrinsic geodesics.)

Theorem D.1 (Track 2 global Fisher-Rao spherical-arc backstop). *Under (H1) and (PI) — for all values of transferred information,*

$$\mathbb{E} d_{FR}(P_{M_{\tau+}|e, M_{\tau-}, G}, P_{M_{\tau+}|e, M_{\tau-}}) \leq 2 \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}.$$

The constant 2 is universal, dimension-free, global — no small-information condition required, no (R), no (K). Tightest form: $\mathbb{E} d_{FR} \leq 2 \arccos(\exp(-I_M/2)) \leq \min(2\sqrt{I_M}, \pi)$, by Jensen on the concave map $\psi(K) := 2 \arccos(\exp(-K/2))$ — ψ is concave on $[0, \infty)$ (verifiable by $\psi'(K) =$

exp($-K/2$)/ $\sqrt{1 - \exp(-K)}$ decreasing). The constant 2 is sharp — approached arbitrarily closely as $N \rightarrow \infty$ on a symmetric N -point family (Section D.1.1 below).

Proof. By Rényi-divergence monotonicity^[30], $D_{1/2}(P\|Q) \leq D_1(P\|Q) = \text{KL}(P\|Q)$, equivalently $\text{BC}(P, Q) := \int \sqrt{pq} \geq \exp(-\text{KL}/2)$. Combined with the chord-arc identity (not inequality) $d_{FR} = 2 \arccos(\text{BC})$ from the unit- L^2 -sphere geometry: $d_{FR} \leq 2 \arccos(\exp(-\text{KL}/2))$. Define $\phi(K) := 4 \arccos^2(\exp(-K/2))$; series expansion at $K = 0$ gives $\phi(K)/K \rightarrow 4$, decreasing thereafter, so $\phi(K) \leq 4K$ globally. Hence $d_{FR}^2 \leq 4 \text{KL}$ slice-wise. Take $\mathbb{E}_G$, substitute the chain rule Lemma 5.1: $\mathbb{E} d_{FR}^2 \leq 4I_M$. Jensen on $\sqrt{x}$: $\mathbb{E} d_{FR} \leq 2\sqrt{I_M}$. The tightest form uses concavity of ϕ before Jensen on $\sqrt{x}$. $\square$

The $\sqrt{2}$ overhead vs. the locally-tight $\sqrt{2}$ — i.e., $2/\sqrt{2} = \sqrt{2}$ in raw constant ratio — is *exactly* the chord-arc factor at the unit L^2 -sphere’s antipode in squared-distance form. Under (PI) alone, the Bhattacharyya-coefficient pseudometric $d_{FR} = 2 \arccos(\text{BC})$ is automatically (PI)-invariant (BC is an f -divergence), and the bound $d_{FR} \leq 2\sqrt{\text{KL}}$ is the cleanest possible (PI)-invariant Fisher-Rao-to-KL bound; (R) + (K) add the local refinement to $\sqrt{2}$. This composition (exact chord-arc identity + Rényi-1/2 monotonicity) is strictly tighter than the (chord-arc inequality + Tsybakov 2.4) composition by factor $\pi/(2\sqrt{2}) \approx 1.11$ — Tsybakov 2.4 is loose by factor 2 at small KL ($\text{Hel}^2 \approx \text{KL}/4$ locally vs Tsybakov’s $\text{Hel}^2 \leq \text{KL}/2$), and Rényi-1/2 monotonicity recovers the lost factor.

D.1.1 Sharpness witness

The constant 2 is sharp via a symmetric N -point construction. Take G uniform on $\{1, \dots, N\}$ with $P_{M_{\tau+}|G=g}$ uniform on the $N-1$ atoms other than g . By symmetry $P_{M_{\tau+}}$ is uniform on all N atoms, so $\text{KL}(P_{M_{\tau+}|G=g}\|P_{M_{\tau+}}) = \log \frac{N}{N-1}$ (constant in g , so $I_M = \log \frac{N}{N-1}$) and $\text{BC} = \sqrt{(N-1)/N}$, giving $d_{FR} = 2 \arcsin(1/\sqrt{N})$ (also constant in g , so $\text{Var}_G(d_{FR}) = 0$ and Jensen is tight). The ratio $\mathbb{E} d_{FR}/\sqrt{I_M} = 2 \arcsin(1/\sqrt{N})/\sqrt{\log \frac{N}{N-1}} \rightarrow 2$ at rate $1 - 1/(12N) + O(N^{-2})$ as $N \rightarrow \infty$. Numerical confirmation: $N = 100$ achieves 99.92% saturation, $N = 1000$ achieves 99.992%. The discrete-alphabet structure — large vocabulary $|V|$ admitting near-disjoint goal-conditional supports — is exactly the structural shape adversarial / jailbreak prompts exploit; the witness is the natural worst-case for goal-coupled architectures.

D.2 Hellinger backstop

The chord companion to Theorem D.1’s arc bound, with the cleanest universal constant.

Theorem D.2 (Hellinger backstop). *Under (H1) and (PI) — for all values of transferred information,*

$$\mathbb{E} \text{Hel}(P_{M_{\tau+}|e, M_{\tau-}, G}, P_{M_{\tau+}|e, M_{\tau-}}) \leq (1/\sqrt{2}) \cdot \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}.$$

Under (H_κ), the architectural factorization recovers: $\mathbb{E} \text{Hel} \leq (1/\sqrt{2}) \sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega | e, M)}$.

Proof. By Tsybakov³¹, Lemma 2.4, $2\text{Hel}^2(P, Q) \leq \text{KL}(P\|Q)$ globally for all probability measures. Apply slice-wise at each $G = g$, take expectation over G , and substitute the chain rule Lemma 5.1. $\square$

Hellinger satisfies (PI) — it is an f -divergence with $f(t) = (\sqrt{t} - 1)^2/2$ preserved under Markov morphisms^[32,33] — and is locally proportional to Fisher-Rao on the unit-sphere representation. Throughout we adopt the *standard statistician’s* Hellinger convention $\text{Hel}^2(P, Q) := \frac{1}{2} \|\sqrt{p} - \sqrt{q}\|_{L^2}^2 = 1 - \int \sqrt{pq}$, so $\text{Hel} \in [0, 1]$, and the *Amari-Nagaoka* Fisher-Rao convention $d_{FR}(P, Q) = 2 \arccos \int \sqrt{pq}$ (twice the great-circle arc on the unit sphere of $\sqrt{p}$ in L^2 , $d_{FR} \in [0, \pi]$). Different conventions in the literature differ by factors of 2; the forms above are the ones consistent with $\text{Hel} \in [0, 1]$, $d_{FR} \in [0, \pi]$, and the $2\text{Hel}^2 \leq \text{KL}$ inequality used in the proof.

The three Fisher-Rao-spine bounds — locally-tight $\sqrt{2}$ (under (H1)+(H4’)+(PI)+(R)+(K)), globally-valid 2 (Theorem D.1, under (H1)+(PI) only), and Hellinger chord $1/\sqrt{2}$ (Theorem D.2, under (H1)+(PI) only) — all operate on the unit L^2 -sphere of $\sqrt{p}$ via the Amari-Nagaoka spherical-arc

convention adopted in Section 3.1. The local theorem under (PI)+(R)+(K)+(H4') coincides with intrinsic Fisher-Rao on parametric submanifolds at second order; the global theorems under (PI) alone bound the *ambient* spherical-arc pseudometric and apply to any pair of probability distributions regardless of submanifold structure. The hypothesis sets are strictly nested and the bounds use the same chord-arc geometry of the L^2 -sphere; the global constants (2 for the arc, $1/\sqrt{2}$ for the chord) are companion bounds at the antipode.

D.3 Hellinger-Fisher-Rao chord-arc geometry

Lemma D.3 (Global chord-arc bound on the unit L^2 -sphere). *For any probability measures P, Q , $d_{FR}(P, Q) \leq \pi \cdot \text{Hel}(P, Q)$, with equality at the antipode $\text{Hel}(P, Q) = 1$ (correspondingly $d_{FR}(P, Q) = \pi$).*

Proof. From $d_{FR} = 4 \arcsin(\text{Hel}/\sqrt{2})$, define $\phi : (0, 1] \rightarrow (0, \pi]$ by $\phi(h) := 4 \arcsin(h/\sqrt{2})/h$, so $d_{FR}/\text{Hel} = \phi(\text{Hel})$. Set $g(h) := 4 \arcsin(h/\sqrt{2})$. Differentiating: $g'(h) = 4/\sqrt{2-h^2}$ for $h \in [0, \sqrt{2})$, $g''(h) = 4h/(2-h^2)^{3/2} > 0$ on $(0, \sqrt{2})$ — so g is *strictly convex* on this interval, with $g(0) = 0$. By the secant-slope monotonicity for convex functions vanishing at zero, $\phi(h) = g(h)/h$ is monotonically increasing on $(0, \sqrt{2})$, with limits $\lim_{h \rightarrow 0^+} \phi(h) = g'(0) = 4/\sqrt{2} = 2\sqrt{2}$ and $\phi(1) = g(1) = 4 \arcsin(1/\sqrt{2}) = \pi$. Hence $\phi(h) \leq \pi$ for all $h \in (0, 1]$, with equality at $h = 1$. $\square$

Local proportionality As $d_{FR} \rightarrow 0$, $\text{Hel} \approx d_{FR}/(2\sqrt{2})$, so Theorem 4.1's Track 2 locally-tight bound $\mathbb{E} d_{FR} \leq \sqrt{2I_M}$ jointly with Lemma D.3 gives $\mathbb{E} \text{Hel} \lesssim (1/2)\sqrt{I_M}$ in the small-information limit — *tighter* than Theorem D.2's global $(1/\sqrt{2})\sqrt{I_M}$ by a factor of $\sqrt{2}$. So in the (H4') regime, Theorem 4.1 + Lemma D.3 jointly give the tightest Hellinger bound; Theorem D.2 (under (PI) only, no (H4')) is the global statement.

D.4 Conjugate-Gaussian Euclidean translation

When (H1) is realized by a parametric family with known Fisher information, the locally-tight Fisher-Rao bound translates to an explicit Euclidean bound on parameter displacement. The conjugate-Gaussian case clarifies the no-go's scope: *Euclidean displacement does not blow up* in the prior-dominant limit, contrary to a naive reading of the chart-rescaling argument.

Theorem D.4 (Conjugate-Gaussian Euclidean bound). *As in Appendix C, the agent uses the conjugate Gaussian prior/likelihood update rule ($\theta \sim \mathcal{N}(0, \tau^2)$, $\Omega \mid \theta \sim \mathcal{N}(\theta, \sigma^2)$), while the operating distribution probing goal-coupling has $\Omega \mid G = g \sim \mathcal{N}(\beta(g), \sigma^2)$. The post-update parameter law under operating G is the Gaussian $\mu_+ \mid G \sim \mathcal{N}(L_{\text{post}}\beta(G), L_{\text{post}}^2\sigma^2)$ with $L_{\text{post}} = \tau^2/(\sigma^2 + \tau^2)$. The locally-tight Fisher-Rao bound translates to the Euclidean bound on random-parameter displacement*

$$\mathbb{E} |\Delta\mu_+|_{\text{Eucl}} \leq L_{\text{post}} \sigma \cdot \sqrt{2I_M} \cdot (1 + o(1)),$$

with prefactor $L_{\text{post}} \sigma = \tau^2 \sigma / (\sigma^2 + \tau^2) \leq \tau/2$ uniformly. Under (H_κ) , $\mathbb{E} |\Delta\mu_+|_{\text{Eucl}} \leq L_{\text{post}} \sigma \sqrt{2\kappa_{\text{processing}}} I(G; \Omega \mid e, M) (1 + o(1))$.

Proof. The post-update parameter law $\mu_+ \mid G$ is Gaussian with mean $L_{\text{post}}\beta(G)$ and variance $L_{\text{post}}^2\sigma^2$ (variance preserved across goals because σ_{post}^2 is fixed by the conjugate-update structure). The Fisher metric on the manifold $\{\mathcal{N}(\mu, L_{\text{post}}^2\sigma^2) : \mu \in \mathbb{R}\}$ in the μ -coordinate is $\mathbf{I}(\mu) = 1/(L_{\text{post}}^2\sigma^2)$, so the Fisher-Rao geodesic distance between two such laws is $d_{FR}(\mu_1, \mu_2) = |\mu_1 - \mu_2|/(L_{\text{post}}\sigma)$ — a flat metric pulled back from $\mathbb{R}$. Hence $|\Delta\mu_+|_{\text{Eucl}} = L_{\text{post}}\sigma \cdot d_{FR}$. Take expectations and substitute the locally-tight Fisher-Rao bound. $\square$

The prefactor $L_{\text{post}}\sigma = \tau^2\sigma/(\sigma^2 + \tau^2)$ is bounded above by $\tau/2$ uniformly (AM-GM at the maximizer $\sigma = \tau$) and *vanishes in both extreme limits*: $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow \infty$ (very noisy observations: the architecture learns essentially nothing from Ω regardless of goal) and $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow 0$ (very sharp observations: $L_{\text{post}} \rightarrow 1$ but $\sigma \rightarrow 0$, so even though the posterior tracks data tightly the data variance vanishes). Maximum at balanced $\sigma = \tau$. The naive “ $\sigma\sqrt{2I}$ ” reading is wrong (correct prefactor is $L_{\text{post}}\sigma$); the “ $\sigma_{\text{post}}\sqrt{2I}$ ” reading is *also* wrong (it conflates inside-posterior covariance

889 $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ with parameter-law variance $L_{\text{post}}^2\sigma^2$). The Section 4.1 no-go still applies as stated to
 890 *charts*: $L_{\text{post}}\sigma$ is family-specific.

891 D.5 Exponential-family Euclidean generalization

892 For an exponential family $P_\theta(x) = h(x) \exp(\theta^\top T(x) - A(\theta))$ with natural parameter $\theta \in \Theta \subseteq \mathbb{R}^d$
 893 and log-partition A , the Fisher information metric is $\mathbf{I}(\theta) = \nabla^2 A(\theta)$. The infinitesimal Fisher-Rao
 894 expansion gives $d_{FR}^2(P_{\theta_1}, P_{\theta_2}) = (\theta_1 - \theta_2)^\top \mathbf{I}(\theta^*)(\theta_1 - \theta_2) + O(\|\theta_1 - \theta_2\|^3)$ for θ^* on the geodesic
 895 between θ_1, θ_2 . So Euclidean displacement on the natural parameter satisfies $\|\theta_1 - \theta_2\|^2 \leq \mathbf{I}_{\min}^{-1} \cdot d_{FR}^2$
 896 where $\mathbf{I}_{\min}$ is the minimum eigenvalue of $\mathbf{I}$ along the geodesic.

897 **Theorem D.5** (Exponential-family Euclidean bound). *For an exponential family in natural-parameter*
 898 *coordinates with Fisher-information minimum-eigenvalue $\mathbf{I}_{\min}$ along the goal-induced geodesic, the*
 899 *locally-tight Fisher-Rao bound translates to*

$$900 \mathbb{E} \|\Delta\theta_{\text{bias}}\|_{\text{Eucl}} \leq \mathbf{I}_{\min}^{-1/2} \cdot \sqrt{2I_M} \cdot (1 + o(1)),$$

$$901 \text{ and under } (H_\kappa): \mathbb{E} \|\Delta\theta_{\text{bias}}\|_{\text{Eucl}} \leq \mathbf{I}_{\min}^{-1/2} \sqrt{2\kappa_{\text{processing}} I(G; \Omega \mid e, M)} (1 + o(1)).$$

902 *Proof.* By Theorem 4.1's Track 2 instantiation, $\mathbb{E} d_{FR}^2 \leq 2I_M(1 + o(1))$. By the infinitesimal
 903 Fisher-Rao expansion, $\|\theta_1 - \theta_2\|^2 \leq \mathbf{I}_{\min}^{-1} d_{FR}^2$. Take expectations and Jensen. $\square$

904 **Worked examples** *Bernoulli on logit chart, restricted to $p \in [\delta, 1 - \delta]$: $\mathbf{I}(\theta) = p(1 - p)$ has no*
 905 *positive lower bound on the full chart $\theta \in \mathbb{R}$; restricting to $p \in [\delta, 1 - \delta]$ gives $\mathbf{I}_{\min} \geq \delta(1 - \delta)$*
 906 *and finite, dimension-free Euclidean bound $\mathbb{E} \|\Delta\theta\|_{\text{Eucl}} \leq [\delta(1 - \delta)]^{-1/2} \sqrt{2I_M}$ — contingent on*
 907 *boundary avoidance. Multinomial on log-ratio chart: Fisher minimum eigenvalue bounded below*
 908 *by $\min_i p_i$ on the simplex interior; for δ -bounded models, $\mathbb{E} \|\Delta\theta\|_{\text{Eucl}} \leq \delta^{-1/2} \sqrt{2I_M}$. Gaussian-*
 909 *mean chart with fixed variance v : $\mathbf{I}(\mu) = 1/v$ constant on the manifold; $\mathbb{E} \|\Delta\mu\|_{\text{Eucl}} \leq \sqrt{v} \sqrt{2I_M}$.*
 910 *Gaussian-scale chart $\theta = \sigma$: $\mathbf{I}(\sigma) = 2/\sigma^2$ — $\mathbf{I}_{\min}^{-1/2} = \sigma/\sqrt{2}$ grows with σ , recovering the no-go's*
 911 *pathology consistent with Section 4.1; reparameterizing to $\theta = \log \sigma$ gives $\mathbf{I}(\log \sigma) = 2$ constant,*
 912 *restoring a finite Euclidean bound.*

913 The pattern: for natural-parameter charts on simplex / discrete-outcome models with the goal-induced
 914 geodesic restricted away from the boundary, the Euclidean Track 2 translation is dimension-free and
 915 finite; for scale-family parametrizations off the natural-parameter chart, the prefactor grows in σ or
 916 λ^{-1} — recovering Section 4.1's pathology consistent with the no-go.

917 E Proofs of main results

918 This appendix collects full proofs deferred from §4–§5 of the main text. Proof structure: each section
 919 restates the formal claim from main text, then provides the full derivation. Auxiliary lemmas used in
 920 proofs are stated where they first arise.

921 E.1 Track 1 cascade — full derivation

922 The umbrella theorem's Track 1 instantiation (Theorem 4.1 under (H1) + (H2')) follows from a
 923 two-step composition: chain rule on the post-update law, then slice-wise Talagrand T_2 .

924 *Proof. Step 1 — Chain rule of relative entropy on the post-update law.* By
 925 Cover and Thomas 2006^[29] Theorem 2.5.3 applied to the conditional law of $M_{\tau+}$ given $(e_\tau, M_{\tau-})$
 926 marginalized over G :

$$927 \mathbb{E}_G [\text{KL}(P_{M_{\tau+} | e_\tau, M_{\tau-}, G} \| P_{M_{\tau+} | e_\tau, M_{\tau-}})] = I(G; M_{\tau+} \mid e_\tau, M_{\tau-}).$$

928 The (H1) standard-Borel regularity makes this hold in the abstract-spaces form (Polyanskiy and Wu²¹,
 929 Theorem 3.4; Gray²², Theorem 5.4). The right-hand side is *transferred* goal-information into the
 930 post-update model state.

931 *Step 2 — Slice-wise Talagrand T_2 .* Apply Hypothesis 4.2 slice-wise at each $G = g$:

$$W_2^2(P_{M_{\tau+}|e,M,G=g}, P_{M_{\tau+}|e,M}) \leq C_{T_2} \cdot \text{KL}(P_{M_{\tau+}|e,M,G=g} \| P_{M_{\tau+}|e,M}).$$

Take expectation over G and substitute Step 1:

$$\mathbb{E}[W_2^2(P_{M_{\tau+}|G}, P_{M_{\tau+}})] \leq C_{T_2} \cdot I(G; M_{\tau+} | e_{\tau}, M_{\tau-}).$$

Jensen's inequality on $\sqrt{x}$ delivers the distance form

$$\mathbb{E} W_2(P_{M_{\tau+}|G}, P_{M_{\tau+}}) \leq \sqrt{C_{T_2}} \cdot \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}. \quad \square$$

The architectural-corollary form (Corollary 4.7 under (H_{κ})) follows by substituting $I(G; M_{\tau+} | e, M_{\tau-}) \leq \kappa_{\text{processing}} \cdot I(G; \Omega_{\tau} | e, M_{\tau-})$ on the right-hand side; the substitution is *definitional* per (4), with the boundedness of $\kappa_{\text{processing}}$ being the substantive content of (H_{κ}) .

E.2 Lipschitz pushforward of T_2 — Stuart-school reduction

Track 1's $(H2')$ is most naturally verified through a two-step Lipschitz-pushforward of T_2 that matches the canonical Stuart-school cascade across the full Strand 2 hypothesis space — not only conjugate-Gaussian. The reduction is one textbook lemma + one elementary composition.

Lemma E.1 (Lipschitz pushforward of T_2). *Let (X, d_X) and (Y, d_Y) be Polish metric spaces, μ on X satisfying Talagrand $T_2(C)$ — that is, $W_2^{d_X}(\nu, \mu)^2 \leq C \text{KL}(\nu \| \mu)$ for all $\nu \ll \mu$ — and let $T : X \rightarrow Y$ be L -Lipschitz. Then $T_*\mu$ on Y satisfies $T_2(L^2C)$.*

Proof. Let $\rho \ll T_*\mu$. Disintegrate $\mu(dx) = \mu_y(dx) T_*\mu(dy)$ along T . The lift $\tilde{\rho}(dx) := \mu_y(dx) \rho(dy)$ satisfies $T_*\tilde{\rho} = \rho$ and $\text{KL}(\tilde{\rho} \| \mu) = \text{KL}(\rho \| T_*\mu)$ (chain-rule decomposition with shared conditionals). For any coupling π of $(\tilde{\rho}, \mu)$, the pushforward $(T \otimes T)_*\pi$ couples $(\rho, T_*\mu)$ with cost $\leq L^2$ times the input cost. Hence $W_2^{d_Y}(\rho, T_*\mu)^2 \leq L^2 W_2^{d_X}(\tilde{\rho}, \mu)^2 \leq L^2 C \text{KL}(\tilde{\rho} \| \mu) = L^2 C \text{KL}(\rho \| T_*\mu)$. (Standard T_2 -tensorization-style argument; cf. Bobkov and Götze²⁸.) $\square$

The reduction needs an additional tensorization lemma — T_2 on the goal-induced parameter measure tensorizes through a Lipschitz-conditional kernel into T_2 on the joint, then projects to the marginal observation law, then pushes forward through the Bayesian-update map.

Lemma E.2 (Tensorization of T_2 with Lipschitz-conditional kernel). *Let P_{θ} on Θ satisfy Talagrand $T_2(C_{\theta})$, and let kernel $K(\theta, \cdot) := P_{\Omega|\theta}$ satisfy (i) $K(\theta, \cdot) \in T_2(C_{\text{slice}})$ uniformly in θ , and (ii) $W_2(K(\theta_1, \cdot), K(\theta_2, \cdot)) \leq L_{\text{lik}} \|\theta_1 - \theta_2\|$. Then the joint $P_{\theta} \otimes K$ on $\Theta \times \Omega$ with the L^2 -product metric satisfies $T_2(C_J)$ with $C_J \leq \max((1 + 2L_{\text{lik}}^2) C_{\theta}, 2C_{\text{slice}})$.*

Proof. Disintegrate $\rho \ll P_{\theta} \otimes K$ as $\rho(d\theta, d\Omega) = \rho_{\theta}(d\Omega) \rho_{\Theta}(d\theta)$. Chain rule on KL: $\text{KL}(\rho \| P_{\theta} \otimes K) = \text{KL}(\rho_{\Theta} \| P_{\theta}) + \int \text{KL}(\rho_{\Omega} \| K(\theta, \cdot)) d\rho_{\Theta}$. Construct a coupling: π^{Θ} optimal W_2 between $(\rho_{\Theta}, P_{\theta})$; for each (θ, θ') , $\pi^{\Omega}(\theta, \theta')$ optimal W_2 between $(\rho_{\theta}, K(\theta', \cdot))$. The product coupling has marginals ρ and $P_{\theta} \otimes K$, with L^2 -product cost $W_2^2(\rho_{\Theta}, P_{\theta}) + \iint W_2^2(\rho_{\theta}, K(\theta', \cdot)) d\pi^{\Theta}$. Triangle + Cauchy-Schwarz with (ii): $W_2^2(\rho_{\theta}, K(\theta', \cdot)) \leq 2W_2^2(\rho_{\theta}, K(\theta, \cdot)) + 2L_{\text{lik}}^2 \|\theta - \theta'\|^2$. Apply (i) to the first piece, $T_2(C_{\theta})$ to the marginal, combine: $W_2^2(\rho, P_{\theta} \otimes K) \leq (1 + 2L_{\text{lik}}^2) C_{\theta} \text{KL}(\rho_{\Theta} \| P_{\theta}) + 2C_{\text{slice}} \int \text{KL}(\rho_{\Omega} \| K(\theta, \cdot)) d\rho_{\Theta} \leq \max((1 + 2L_{\text{lik}}^2) C_{\theta}, 2C_{\text{slice}}) \text{KL}(\rho \| P_{\theta} \otimes K)$. $\square$

Theorem E.3 (Stuart-school reduction — generalized cascade-constant transfer). *Suppose:*

(C0) (Goal-induced parameter concentration.) $P_{\theta} := \theta_* P_G$ satisfies Talagrand $T_2(C_{\theta})$ on parameter space.

(C1) (Class 1 architecture.) $G \rightarrow \Omega_{\tau} \rightarrow M_{\tau+}$ is a Markov chain conditional on $(e_{\tau}, M_{\tau-})$.

(C2) (Deterministic Bayesian update.) $M_{\tau+} = \mu^{\Omega_{\tau}}$ for a measurable map $\Omega \mapsto \mu^{\Omega}$ on $\text{supp } P_{\Omega}$.

(C3) (Stuart-school Lipschitz posterior.) $W_2(\mu^{\Omega}, \mu^{\Omega'}) \leq L_{\text{post}} \|\Omega - \Omega'\|$ uniformly on $\text{supp } P_{\Omega}$.

(C4a) (Slice T_2 .) For P_{θ} -a.e. θ , $P_{\Omega|\theta} \in T_2(2/\rho_{\text{noise}})$.

(C4b) (Lipschitz-in- θ kernel.) $W_2(P_{\Omega|\theta_1}, P_{\Omega|\theta_2}) \leq L_{\text{lik}} \|\theta_1 - \theta_2\|$.

Then:

975 (a) (H2') holds on the goal-marginal post-update model law with $C_{T_2} \leq L_{\text{post}}^2 \max((1 +$
 976 $2L_{\text{lik}}^2) C_\theta, 4/\rho_{\text{noise}})$.

977 (b) Track 1 yields $\mathbb{E} W_2^2(P_{M_{\tau+}|G}, P_{M_{\tau+}}) \leq C_{T_2} I(G; M_{\tau+} | e_\tau, M_{\tau-})$.

978 (c) The cascade constant has the canonical Stuart-school form $\propto L_{\text{post}}^2/\rho_{\text{noise}}$ across the Stuart-school
 979 hypothesis space, with an absolute multiplicative overhead from the kernel-tensorization step. The
 980 tight canonical constant $2L_{\text{post}}^2/\rho_{\text{noise}}$ is recovered on instances where the post-update parameter law
 981 admits a direct sub-Gaussian read-off (conjugate-Gaussian and exponential-family-conjugate cases
 982 — see Appendix C for the worked-through Class 1 instance).

983 *Proof.* By Lemma E.2 applied to (C0)+(C4a)+(C4b), the joint $P_\theta \otimes K \in T_2(C_J)$ on $\Theta \times \Omega$ with
 984 $C_J \leq \max((1 + 2L_{\text{lik}}^2)C_\theta, 4/\rho_{\text{noise}})$. The projection $(\theta, \Omega) \mapsto \Omega$ is 1-Lipschitz, so by Lemma E.1,
 985 $P_\Omega \in T_2(C_J)$. The map $\Omega \mapsto \mu^\Omega$ is L_{post} -Lipschitz by (C3), so by Lemma E.1 again, $P_{M_{\tau+}|e, M_{\tau-}} \in$
 986 $T_2(L_{\text{post}}^2 C_J) = T_2(C_{T_2})$. (b) follows from Track 1's standard cascade (Appendix E.1). (c) is the
 987 matching of canonical form; the tight constant on conjugate instances comes from a *direct* sub-
 988 Gaussian read-off on the post-update parameter law (the structural Lipschitz-composition route is
 989 blind to the Bayesian-update contraction). $\square$

990 Each Strand 2 paper — Stuart 2010^[7], Sprungk 2020^[8], Cvetković and Lie 2025^[9],
 991 Dolera and Mainini 2023^[10], Garbuno-Iñigo et al. 2023^[11], Lie et al. 2018^[34],
 992 Hosseini et al. 2025^[12] — establishes a form of (C3) under increasingly weak hypotheses.
 993 Theorem E.3 is a structural slot they all fit into via the unbundled (C0)+(C3)+(C4a)+(C4b) hypothesis
 994 set; anywhere the Stuart-school cascade applies, Track 1 specializes to a constant of the canonical
 995 form $\propto L_{\text{post}}^2/\rho_{\text{noise}}$. The result is *not* a sub-case statement — Stuart-school proves outputs (matching
 996 lower bounds, multi-metric extensions, sub-Lipschitz refinements, contraction rates, randomized
 997 forward models) outside Track 1's reach. The averaged-over-goal form of Track 1 also cannot recover
 998 Stuart-school's pointwise pair-distance information on its own; the two literatures share machinery
 999 and exchange a constant of the same form, not theorem closures. The averaged-vs-pointwise gap and
 1000 the structural-vs-direct-sub-Gaussian gap on constant tightness are the two precise failure modes of
 1001 the alternative reading “Track 1 contains the Stuart-school lineage as a strict sub-case.”

1002 E.3 Track 2 cascade — full derivation

1003 The umbrella theorem's Track 2 instantiation (under (H1) + (H4') + (PI) + (R) + (K)) follows in one
 1004 step using the KL-to-Fisher-Rao expansion.

1005 **Lemma E.4** (KL-to-Fisher-Rao second-order expansion). *On the ambient L^2 -sphere of $\sqrt{p}$ equipped*
 1006 *with the Fisher metric $\mathbf{I}$ (the maximal statistical manifold, of which every probability distribution is a*
 1007 *point), for nearby P, Q : $\text{KL}(P \| Q) = \frac{1}{2} d_{FR}^2(P, Q) + O(d_{FR}^3)$ in the Amari-Nagaoka spherical-arc*
 1008 *convention.*

1009 **Lemma E.5** (Uniform third-order remainder under (H4')). *Suppose the ambient statistical manifold*
 1010 *along the goal-induced neighborhood admits a chart θ in which (i) the Amari-Chentsov tensor satisfies*
 1011 *$|T_{ijk}(\theta^*) \delta^i \delta^j \delta^k| \leq \mathcal{T}_* \|\delta\|^3$ for all θ^* on the geodesic between θ and $\theta + \delta$, and (ii) the Fisher matrix*
 1012 *is uniformly bounded below: $\mathbf{I}(\theta^*) \succeq \mathbf{I}_{\min} \cdot \text{Id}$ with $\mathbf{I}_{\min} > 0$, on the same neighborhood. Then for*
 1013 *every pair (P_g, P_*) on the neighborhood with $\text{KL}(P_g \| P_*) \leq \delta_* \leq \mathbf{I}_{\min}^3/(8\mathcal{T}_*^2)$,*

$$1014 \quad d_{FR}^2(P_g, P_*) \leq 2 \text{KL}(P_g \| P_*) \cdot (1 + R_3(\delta_*)), \quad R_3(\delta_*) := \frac{2\mathcal{T}_* \sqrt{2\delta_* / \mathbf{I}_{\min}^3}}{1 - \mathcal{T}_* \sqrt{2\delta_* / \mathbf{I}_{\min}^3}} \xrightarrow{\delta_* \rightarrow 0} 0.$$

1015 *Proof. (of Lemma E.5.)* The Lagrange-form Taylor remainder for the KL expansion (Amari and
 1016 Nagaoka¹⁹ §3.7 Theorem 3.1) gives, with hypothesis (i),

$$1017 \quad |\text{KL}(P_g \| P_*) - \frac{1}{2} \delta^\top \mathbf{I}(\theta^*) \delta| \leq \frac{1}{6} \mathcal{T}_* \|\delta\|^3,$$

1018 for some θ^* on the geodesic. Hypothesis (ii) gives $\delta^\top \mathbf{I}(\theta^*) \delta = d_{FR}^2(P_g, P_*) (1 + O(\|\delta\|))$ at leading
 1019 order with the leading-order remainder also controlled by $\mathcal{T}_*/\mathbf{I}_{\min}^{3/2}$; folding both into a single inequality
 1020 and using $d_{FR}^2 \geq \mathbf{I}_{\min} \|\delta\|^2$ (so $\|\delta\| \leq d_{FR}/\sqrt{\mathbf{I}_{\min}}$, $\|\delta\|^3 \leq d_{FR}^3/\mathbf{I}_{\min}^{3/2}$),

$$1021 \quad d_{FR}^2(P_g, P_*) \leq 2 \text{KL}(P_g \| P_*) + \frac{1}{3} \mathcal{T}_* d_{FR}^3/\mathbf{I}_{\min}^{3/2}.$$

Setting $u := d_{FR}(P_g, P_*) \geq 0$ and $K := \text{KL}(P_g \| P_*) \leq \delta_*$, this is $u^2 \leq 2K + (\mathcal{T}_*/3)u^3/\mathbf{I}_{\min}^{3/2}$, i.e.,
 $u^2(1 - (\mathcal{T}_*/3)u/\mathbf{I}_{\min}^{3/2}) \leq 2K$. The polynomial $f(u) := u^2 - (\mathcal{T}_*/3)u^3/\mathbf{I}_{\min}^{3/2} - 2K$ is increasing on
 $[0, 2\mathbf{I}_{\min}^{3/2}/\mathcal{T}_*]$ (where $f'(u) = u(2 - \mathcal{T}_*u/\mathbf{I}_{\min}^{3/2}) \geq 0$) and has a unique root $u^*(K)$ in that interval,
dominating u . Since $f(\sqrt{2K}) = -(\mathcal{T}_*/3)(2K)^{3/2}/\mathbf{I}_{\min}^{3/2} < 0$, the root satisfies $u^*(K) \geq \sqrt{2K}$; the
explicit upper bound from substituting $u \leq u^*(K)$ back into $u^2 \leq 2K + (\mathcal{T}_*/3)u^3/\mathbf{I}_{\min}^{3/2}$ and iterating
once gives $u^*(K)^2 \leq 2K(1 + R_3(K))$ with $R_3(K) = (2\mathcal{T}_*/3)\sqrt{2K}/\mathbf{I}_{\min}^{3/2}/(1 - (\mathcal{T}_*/3)\sqrt{2K}/\mathbf{I}_{\min}^{3/2})$
for $K \leq \mathbf{I}_{\min}^3/(8\mathcal{T}_*^2)$ (which keeps the denominator $\geq 2/3$). Hence $u^2 \leq 2K(1 + R_3(K)) \leq$
 $2K(1 + R_3(\delta_*))$, and $R_3(\delta_*) \rightarrow 0$ as $\delta_* \rightarrow 0$. $\square$

Proof. (Track 2 cascade.) Apply Lemma E.5 under (H1) + (H4') slice-wise at each $G = g$:

$$d_{FR}^2(P_{M_{\tau+}|e,M,G=g}, P_{M_{\tau+}|e,M}) \leq 2 \text{KL}(P_{M_{\tau+}|G=g} \| P_{M_{\tau+}}) (1 + R_3(\delta_*)),$$

with $R_3(\delta_*) \rightarrow 0$ as $\delta_* \rightarrow 0$. The (H4') regime supplies the slice-wise $\text{KL}_g \leq \delta_*$ premise; the
chart-existence premise (uniform Amari-Chentsov bound + minimum Fisher eigenvalue) is ambient-
Fisher-curvature regularity along the goal-induced neighborhood — implied by (H4') on smooth
statistical-manifold-sub-case (H1) families since both quantities depend continuously on the operating
point and the neighborhood is compactly generated.

Take expectation over G and substitute the chain-rule identity from Lemma 5.1:

$$\mathbb{E}[d_{FR}^2(P_{M_{\tau+}|G}, P_{M_{\tau+}})] \leq 2 \cdot I(G; M_{\tau+} | e_{\tau}, M_{\tau-}) \cdot (1 + o(1)).$$

Jensen on $\sqrt{x}$ delivers the distance form

$$\mathbb{E} d_{FR}(P_{M_{\tau+}|G}, P_{M_{\tau+}}) \leq \sqrt{2} \cdot \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})} \cdot (1 + o(1)). \quad \square$$

The constant $\sqrt{2}$ emerges from the second-order coefficient $\frac{1}{2}$ in the KL-to-Fisher expansion (Amari
and Nagaoka¹⁹ §3.7 Theorem 3.1) — exactly what (K) crystallizes as a normalization. Outside (H4'),
the locally-tight bound's $(1 + o(1))$ remainder is no longer uniformly sharp; companion bounds in
Appendix D cover the moderate-to-large I_M regime under strictly weaker hypotheses (Track 2 global,
Hellinger backstop).

E.4 No-go on Euclidean chart norms — full proof

Theorem 4.4's proof uses a chart-rescaling lemma plus a one-line consequence.

Lemma E.6 (Chart-rescaling sensitivity of Euclidean displacement). *Let $\mathcal{M}$ be a statistical manifold
and $\phi : U \rightarrow \mathbb{R}^d$ a chart on an open set $U \subseteq \mathcal{M}$. For any $a > 0$, write $\phi_a := a\phi$ for the
rescaled chart. For every pair of probability laws μ, ν on U (in particular, the goal-conditional
and goal-marginal post-update laws of (3)): (a) The chart-Euclidean Wasserstein distance scales
linearly: $W_2^{\phi_a}(\mu, \nu) = a \cdot W_2^{\phi}(\mu, \nu)$. (b) Conditional KL, conditional mutual information, Fisher-Rao
spherical-arc distance, and Hellinger distance are all chart-invariant: they take the same value under
 ϕ and ϕ_a .*

Proof. (a) The pushforward under the linear map $x \mapsto ax$ rescales Euclidean distances by a , hence
rescales any optimal coupling's transport cost. (b) KL depends only on the underlying densities
and dominating measure (Radon-Nikodym derivatives are reparameterization-invariant). The Fisher
metric tensor transforms covariantly under chart change, so the integrated geodesic distance is invariant.
Hellinger as an f -divergence is reparameterization-invariant. $\square$

Proof. Proof of Theorem 4.4. Suppose for contradiction that a coordinate-independent $C_0 < \infty$
exists such that for every chart ϕ on $\mathcal{M}$ and every architecture satisfying (H1)+(H2') with non-zero
post-update displacement,

$$\|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi} \leq C_0 \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}.$$

For any architecture realizing $W_2^{\phi}(P_{M_{\tau+}|G,e,M_{\tau-}}, P_{M_{\tau+}|e,M_{\tau-}}) > 0$ and finite positive transferred
information, applying the supposed bound under the rescaled chart ϕ_a gives, by Lemma E.6,

$$a \cdot \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi} = \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi_\alpha} \leq C_0 \sqrt{I},$$

with the right-hand side unchanged across charts. Taking $a \rightarrow \infty$ contradicts the fixed $C_0 \sqrt{I}$. The same conclusion holds for the architectural-corollary form $C \sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega \mid e, M)}$ under (H_κ) : both sides inherit Lemma E.6’s chart-(non)invariance, and the contradiction goes through verbatim. $\square$

The proof avoids type-shifts between Dirac points and pushforward laws, never invokes $(H2')$ -violating two-point constructions, and does not assert any specific value of I — it only uses that some architecture realizes finite positive W_2^ϕ and finite positive I , automatic on any non-trivial goal-coupled architecture (e.g., the conjugate-Gaussian setup with any $\sigma, \tau > 0$ and a non-degenerate continuous goal distribution). The Gaussian scale family with two charts (Chart A: σ ; Chart B: $\log \sigma$) is the canonical illustration: same intrinsic geometry, Chart-A Euclidean displacement grows linearly while Chart-B stays bounded as the operating scale grows.

E.5 Čencov uniqueness and sharpness — full proof

Theorem 4.5 has two parts.

Proof. Proof of (a) — Uniqueness of the metric. (PI) + (R) invokes Čencov’s uniqueness theorem^[18] (modern treatment Ay et al.³⁵, Theorem 5.1) on the *categorical metric assignment* of (PI): the assignment must give the Fisher information metric $\mathbf{I}$ on every standard parametric statistical model, up to a single global positive scalar $c > 0$. The induced metric on the model family at hand satisfies $d_{\mathcal{M}} = \sqrt{c} d_{FR}$ where d_{FR} is at the Amari-Nagaoka normalization $\mathbf{I}(\theta) = -\mathbb{E}[\nabla^2 \log p_\theta]$. The KL-to-Fisher second-order expansion (Amari and Nagaoka¹⁹ §3.7 Theorem 3.1) gives $\text{KL}(P_\theta \| P_{\theta+\delta}) = \frac{1}{2} d_{FR}^2 + O(d_{FR}^3)$, so $\frac{1}{2} d_{\mathcal{M}}^2 = (c/2) d_{FR}^2 + O(d_{FR}^3)$. Constraint (K) requires $\frac{1}{2} d_{\mathcal{M}}^2$ agree with KL at second order, forcing $c = 1$ and $d_{\mathcal{M}} = d_{FR}$. $\square$

Proof. Proof of (b) — Sharpness of the constant. Use the symmetric two-point witness on the conjugate-Gaussian Class 1 (Separated) family. Take $G \in \{-a, +a\}$ uniform with $\beta(g) = g$, prior $\theta \sim \mathcal{N}(0, \tau^2)$, observation $\Omega \mid G = g \sim \mathcal{N}(g, \sigma^2)$, and posterior $M_{\tau+} = L_{\text{post}} \Omega$. Then $P_{M_{\tau+} | G=g} = \mathcal{N}(L_{\text{post}} g, L_{\text{post}}^2 \sigma^2)$ and the goal-marginal is the equal-weight mixture of these two Gaussians (centered at $\pm L_{\text{post}} a$). For fixed σ and $a/\sigma \rightarrow 0$ the mixture’s variance excess is $L_{\text{post}}^2 a^2 = O((a/\sigma)^2) \cdot \sigma^2$ — second-order in a/σ — while the mean-shift is first order in a/σ , so the mixture is approximately Gaussian at $\mathcal{N}(0, L_{\text{post}}^2 \sigma^2)$ to leading order. The Fisher-Rao distance between two equal-variance univariate Gaussians is $|\mu_1 - \mu_2|/\sigma_{\text{shared}}$, giving $d_{FR}(P_{M_{\tau+} | G=\pm a}, P_{M_{\tau+}}) = a/\sigma$ at leading order, *exactly equal across both goal slices by symmetry*. Hence $\text{Var}_G(d_{FR}) = O((a/\sigma)^4)$ and Jensen is tight in the small-SNR limit. The transferred information for binary-Gaussian channels at small SNR is $I_M = (a/\sigma)^2/2 + O((a/\sigma)^4)$. Therefore $\mathbb{E} d_{FR}/\sqrt{I_M} = (a/\sigma)/\sqrt{(a/\sigma)^2/2} \cdot (1 + o(1)) = \sqrt{2} (1 + o(1))$ as $a/\sigma \rightarrow 0$, and equivalently $\mathbb{E} d_{FR}^2/I_M \rightarrow 2$. The sharpness holds at the first-moment level — Jensen is asymptotically tight by the symmetry-induced near-constancy of d_{FR} across slices — so any candidate constant $C < \sqrt{2}$ fails on this family. Combined with (a), $C = \sqrt{2}$ is the unique sharp upper-bound constant. $\square$

The upshot: once we adopt (PI) at full Markov-morphism strength, plus Riemannian structure, plus KL as the cascade’s information coordinate, the *only* such-invariant choice of norm on $\mathcal{M}$ is Fisher-Rao, and the *unique sharp* upper-bound constant is $\sqrt{2}$. Alternative chart-independent commitments — TV with Pinsker, Hellinger-as-divergence, χ^2 , Rényi — give weaker bounds: TV-Pinsker is non-tight (the constant is universal but the inequality is loose except at coincident measures); Hellinger-as-divergence and χ^2 -as-divergence violate (R); switching divergences on the RHS violates (K). The global Theorem D.1’s $C = 2$ is also sharp in its own (PI)-only regime — approached arbitrarily closely as $N \rightarrow \infty$ on the symmetric N -point family of Section D.1.1.

E.6 Coupled-class autoregressive connectivity — full statement and proof

Lemma E.7 (Coupled-class autoregressive connectivity). *Let $\mathcal{G}_\theta$ be the directed computational graph of an autoregressive sequence model with parameters θ . Suppose at every layer $\ell \geq 1$ each*

per-position update factors as $h_\ell^{(j)} = h_{\ell-1}^{(j)} + F_\ell^{(j)}(\{h_{\ell-1}^{(i)}\}_{i \leq j}; \theta) + G_\ell(h_{\ell-1}^{(j)}; \theta)$, with $F_\ell^{(j)}$ a causally-aggregated cross-position mixer and G_ℓ position-wise. If goal positions i_G and evidence positions i_E satisfy $i_G \cap i_E = \emptyset$, then for every layer $\ell \geq 1$ and position $j \geq \max(i_G \cup i_E)$, $h_\ell^{(j)}$ has a directed-graph path back to every $i_G \leq j$ whenever the per-source non-degeneracy condition $(\star)$ holds: for each $i \leq j$, $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i)}$ is generically non-zero on the operating set.

Corollary E.8 (Named-architecture instantiations). *Per-source non-degeneracy is satisfied generically by: (a) plain decoder-only transformer attention (non-degenerate softmax attention weights, non-singular value projection); (b) linear-attention transformers^[36] with non-vanishing feature map (e.g., $\phi = \text{elu} + 1$, $\phi = \exp$); (c) selective state-space models / Mamba^[37] (non-singular discretized $\bar{A} = \exp(\Delta A)$ with $A < 0$ diagonal, non-zero B and C); (d) RWKV time-mixing^[38] (positive time-decay); (e) RetNet^[39] (decay $\gamma \in (0, 1)$); (f) long-convolution / Hyena architectures^[40] (full-support causal kernel).*

Each is structurally Class 3 (Coupled) under standard parameterization.

The proof is a one-induction over layer depth, structurally identical across the architectures of Corollary E.8 — the per-source non-degeneracy condition is the architecture-specific input, the directed-graph-path conclusion is the architecture-agnostic output. Robust to RMSNorm^[41] / Group-Norm^[42] / FlashAttention^[43] (position-wise / mathematically-equivalent), causal masking (preserves $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i_G)} \neq 0$ for $i_G \leq j$), and sliding-window or sparse attention (preserves connectivity by composition across layers when the window pattern admits a multi-hop path). Architectures with explicit goal-to-output blocking — token-dropping at intermediate layers, hard-routing MoE-attention with goal-conditional partitioning, or strictly goal-blind retrieval components — fall outside the lemma’s scope and are appropriately Class 2 (Partial).

Proof. Proof of Lemma E.7. By induction on layer depth, in the unified causally-aggregated-mixer form.

Layer 0 (embedding). $h_0^{(i)}$ depends only on X_i . Goal embeddings $h_0^{(i_G)}$ are functions only of the goal tokens at those positions; same for evidence positions i_E . No cross-position dependence yet.

Layer $\ell \geq 1$. The per-position update $h_\ell^{(j)} = h_{\ell-1}^{(j)} + F_\ell^{(j)}(\{h_{\ell-1}^{(i)}\}_{i \leq j}; \theta) + G_\ell(h_{\ell-1}^{(j)}; \theta)$ induces a directed position-graph with two edge types: (i) *same-position* residual edge $h_{\ell-1}^{(j)} \rightarrow h_\ell^{(j)}$ always present; (ii) *cross-position* mixer edges $h_{\ell-1}^{(i)} \rightarrow h_\ell^{(j)}$ present whenever $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i)} \neq 0$, i.e., whenever $(\star)$ holds at (j, i) .

Inductive claim. For every $\ell \geq 0$ and every input position $i \leq j$ along which $(\star)$ holds at some intermediate layer, $h_\ell^{(j)}$ has a directed-graph path back to input position i .

Inductive step. The same-position residual chain gives $i = j$. For $i \neq j$ with $i \leq j$, $(\star)$ at any single layer $\ell' \leq \ell$ along the path gives a cross-position edge $h_{\ell'-1}^{(i)} \rightarrow h_{\ell'}^{(j)}$, which composes with the residual chain on either side to yield a directed-graph path from input position i to $h_\ell^{(j)}$.

Coupled-class conclusion. For every goal position $i_G \leq j$, $h_\ell^{(j)}$ has a path back to position i_G whenever $(\star)$ holds at (j, i_G) at some layer. So G (encoded at positions i_G) is causally upstream of every quantity contributing to a post-update model state read off from position $j \geq \max(i_G \cup i_E)$. Class 3 (Coupled) by construction. $\square$

Per-architecture verifications of $(\star)$ Each architecture in Corollary E.8 satisfies $(\star)$ generically; the verifications are short and architecture-specific.

(a) Self-attention. $F_\ell^{(j)} = \sum_i \alpha_{ji}^{(\ell)}(h_{\ell-1}) V_\ell h_{\ell-1}^{(i)}$ with softmax weights $\alpha_{ji} > 0$. Per-source partial $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i)} = \alpha_{ji}^{(\ell)} V_\ell + \text{softmax-derivative cross-terms}$. Strictly positive softmax + generically non-trivial value-projection action ($V_\ell h_{\ell-1}^{(i)} \neq 0$ on the operating set, or equivalently V_ℓ has no left-null subspace containing the operating-set hidden states) $\Rightarrow (\star)$.

1159 (b) *Linear attention*^[36]. $F_\ell^{(j)}$ is a kernel-weighted sum with per-source factor
 1160 $\phi(Kh^{(i)})^\top \phi(Qh^{(j)})Vh^{(i)}$. $(\star)$ holds when ϕ is non-vanishing on the operating set: standard choices
 1161 $\phi = \text{elu} + 1$, $\phi = \exp$ are strictly positive.

1162 (c) *Selective state-space models / Mamba*. Unrolling the recurrence: $y_\ell^{(j)} =$
 1163 $C_\ell^{(j)} \sum_{i \leq j} (\prod_{k=i+1}^j \bar{A}_\ell^{(k)}) \bar{B}_\ell^{(i)} x_\ell^{(i)}$. Per-source partial leading factor $C_\ell^{(j)} \prod \bar{A} \cdot \bar{B}_\ell^{(i)}$. Under standard
 1164 parameterization $\bar{A} = \exp(\Delta A)$ with $A < 0$ diagonal (HiPPO-style) and $\Delta = \text{softplus}(\cdot) > 0$,
 1165 eigenvalues lie in $(0, 1)$ — non-singular for every k . $\bar{B}$ and C generically non-zero. Selectivity
 1166 is *gain modulation* in a one-parameter family, not a path-cutting routing decision; quantitative
 1167 attenuation lives in $\kappa_{\text{processing}}$.

1168 (d) *RWKV*. Time-mixing $\text{wkv}_t = \frac{\sum_{i \leq t} e^{-(t-1-i)w+k_i} v_i + e^{u+k_t} w_t}{\sum_{i < t} e^{-(t-1-i)w+k_i} + e^{u+k_t}}$ has strictly-positive per-source weight
 1169 $e^{-(t-1-i)w+k_i}$. $(\star)$ holds.

1170 (e) *RetNet*. $S_n = \gamma S_{n-1} + k_n^\top v_n$, $y_n = q_n S_n$ with $\gamma \in (0, 1)$ unrolls to $y_n = q_n \sum_{i \leq n} \gamma^{n-i} k_i^\top v_i$.
 1171 Per-source partial $q_n \gamma^{n-i} k_i^\top$ generically non-zero.

1172 (f) *Long-convolution / Hyena*. $y_j = \sum_{i \leq j} h_{j-i} x_i$. Per-source partial is the kernel value h_{j-i} ,
 1173 non-zero under full causal support.

1174 *Robustness to common variants*. RMSNorm^[41], GroupNorm^[42], and FlashAttention^[43] preserve the
 1175 position-graph (position-wise normalization or mathematically-equivalent attention). Causal masking
 1176 preserves $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i_G)} \neq 0$ for $i_G \leq j$. Sliding-window or sparse attention preserves connectivity
 1177 by multi-hop composition when the window pattern admits a path.

1178 *Class 2 boundary*. Hard-routing MoE-attention with goal-conditional partitioning and token-dropping
 1179 at intermediate layers fall outside $(\star)$ structurally — the goal-to-output graph path is severed by
 1180 hypothesis. Vanishing-feature-map linear attention ($\phi(x) = \text{ReLU}(x)$ on negative inputs) and SSM
 1181 near-state-collapse (Δ large but finite) are *quantitative* boundaries: the directed-graph path remains,
 1182 $(\star)$'s genericity holds, and structural Class 3 is intact, but $\kappa_{\text{processing}}$ is small. *Exact* state-collapse
 1183 ($\bar{A} \rightarrow 0$, $\Delta \rightarrow \infty$) is the limiting case where the cross-position partial $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i)} \rightarrow 0$ exits $(\star)$'s
 1184 genericity — beyond the lemma's quantitative certification but reachable only in the limit, not for any
 1185 finite-magnitude Δ . Strictly goal-blind retrieval components yield system-level Class 2.

1186 Lemma E.7 establishes a *downstream-output graph-reachability* claim: at every layer and every
 1187 position causally downstream of the goal and evidence under standard causal masking, the activation
 1188 has a directed-graph path back to at least one goal position. *Three caveats*:

- 1189 • *Graph reachability is not the same as causal-influence magnitude*. A directed-graph path
 1190 with near-zero attention weights produces a formal edge with negligible quantitative influence
 1191 on $h_\ell^{(j)}$. The connectivity claim is therefore *structural* — about the existence of a path,
 1192 not the magnitude of effect — and translates into Class 3 (Coupled) classification under
 1193 the Markov-blanket conditional-independence reading. Bounding $\kappa_{\text{processing}}$ requires the
 1194 architectural-bandwidth conditions of Appendix B.3 (information-bottleneck on the goal
 1195 channel; bounded direct-channel capacity uniformly over operating distributions).
- 1196 • *Causal masking restricts the lemma's scope to downstream positions*. For $j < \min(i_G)$ —
 1197 positions strictly upstream of any goal token — there is no causal path from goal to $h_\ell^{(j)}$
 1198 under causal attention masks; those positions are Class 1 (Separated) in their own right.
- 1199 • *Sparse and sliding-window attention require a multi-hop reachability assumption*. The
 1200 connectivity proof composes single-layer attention edges across layers; for sliding-window
 1201 or sparse-attention architectures, multi-hop connectivity from i_G to j requires at least one
 1202 composition of single-layer paths through the window/sparsity pattern. Architectures whose
 1203 sparsity pattern excludes goal-to-output communication entirely fall outside the lemma's
 1204 hypothesis and are Class 2 (Partial).

1205 The bias-bound *interpretation* — bounding the displacement of an implicit posterior under goal-
 1206 conditional re-prompting — additionally invokes the in-context-learning correspondence^[13–16] be-
 1207 tween the next-token distribution and an implicit posterior. The structural Coupled-class claim of
 1208 Lemma E.7 is robust without it; the bias-bound *application* is conditional on the in-context-learning

1209 correspondence, which has its own scope (in-distribution prompts: tight; OOD/jailbreak: degraded to
1210 a bound on output-distribution displacement rather than belief-state displacement).

1211 **F Extended related work**

1212 §2 Related Work names the two main lineages this paper composes (LLM hallucination theory;
1213 Bayesian inverse-problems posterior stability). This appendix covers the additional adjacent strands —
1214 Fisher-Rao Bayesian sensitivity, Owhadi-Scovel-Sullivan brittleness, mean-field variational inference
1215 stability, and the Markov-blanket apparatus from active inference — that sit close enough to require
1216 explicit cite-and-distinguish but do not resolve the goal-coupling-displacement gap.

1217 **F.1 Strand 1: LLM hallucination theory**

1218 The recent statistical-inevitability and capacity-impossibility lineage targets the *frequency* of hallucina-
1219 tion on architecture-agnostic grounds. Kalai and Vempala 2024^[2] establish that calibrated language
1220 models must hallucinate at a rate close to the fraction of singleton facts in the training data — a
1221 statistical lower bound that holds independently of the transformer architecture. Kalai et al. 2026^[3]
1222 extend this to argue that hallucinations originate as binary-classification errors compounded by
1223 training-and-evaluation incentives that reward guessing under uncertainty. Karbasi et al. 2025^[4]
1224 reduce hallucination *detection* to language identification in the Gold-Angluin framework, showing
1225 that detection is impossible from positive examples alone but feasible with expert-labeled negatives.
1226 Wu et al. 2025^[5] establish a VC-dimension-based no-free-lunch result for non-hallucinating learning.
1227 Suzuki et al. 2025^[6] give a probabilistic positive result — hallucinations can be made statistically neg-
1228 ligible — counterbalancing the computability-theoretic inevitability arguments. Guo and Li 2026^[44]
1229 frame hallucination as a rate-distortion theorem for membership testing under capacity constraints.
1230 Liu et al. 2025^[45] formalize hallucinations as estimation failures with high-probability lower bounds
1231 on the hallucinate rate. Chlon et al. 2025^[46] frame transformers as Bayesian-in-expectation-not-
1232 realization with $O(\log n)$ permutation dispersion. Zeng et al. 2026^[47] decompose hallucination risk
1233 into data-driven and reasoning-driven components.

1234 Our distance from this strand: we bound a *displacement*, not a frequency, under named *geometric*
1235 assumptions, and we engage the *architecture’s coupling structure* directly via $\kappa_{\text{processing}}$. The bound is
1236 not a substitute for the frequency bounds; it is an orthogonal axis. Among the recent papers, Zeng
1237 et al.’s decomposition is closest in shape to our $\kappa \times I$ factorization but differs on structural axis
1238 (training-time vs inference-time, vs our architectural-vs-informational).

1239 **F.2 Strand 2: Bayesian inverse-problems posterior stability**

1240 The transport-inequality and Lipschitz-posterior machinery driving Track 1 is mature. Stuart 2010^[7]
1241 establishes well-posedness of Bayesian inverse problems on function spaces, with Lipschitz continuity
1242 of the posterior in the data. Sprungk 2020^[8] extends this to local Lipschitz stability with respect
1243 to perturbations of the prior and the log-likelihood, in TV / Hellinger / Wasserstein / KL distances.
1244 Cvetković and Lie 2025^[9] match the upper bounds with lower bounds, emphasizing that posterior
1245 sensitivity *increases* as the posterior concentrates. Dolera and Mainini 2023^[10] obtain Lipschitz-
1246 continuity of probability kernels in the optimal-transport framework, with explicit constants in
1247 terms of Fisher-information functionals and weighted Poincaré constants — the closest-in-spirit
1248 prior result to Track 1’s form. Garbuno-Iñigo et al. 2023^[11] develop integral-probability-metric
1249 perturbation analysis under log-Sobolev-style functional inequalities. Latz 2020^[48] provides well-
1250 posedness under weaker continuity. Lie et al. 2018^[34] give Hellinger-distance bounds for randomized
1251 forward models. Hosseini et al. 2025^[12] develop conditional optimal transport on function spaces
1252 for amortized Bayesian inference, with explicit regularity estimates on conditioning maps from prior
1253 to posterior — the closest single-paper match for Track 1’s cascade. Dolera et al. 2024^[49] derive
1254 Wasserstein-based posterior contraction rates. Del Moral 2025^[50] gives Sinkhorn / Schrödinger
1255 bridge contraction via log-Sobolev — most recent transport-inequality-plus-LSI work, adjacent to
1256 Track 1’s machinery.

1257 Each of these results establishes a form of Lipschitz-in-data continuity of the posterior under in-
1258 creasingly weak hypotheses. Theorem E.3 shows that any such Lipschitz-posterior result, combined
1259 with T_2 concentration on the goal-induced parameter measure and slice-wise sub-Gaussianity of the

conditional data law, is a sufficient verification path for Track 1’s (H2’) on the post-update model law in any Class 1 (Separated) architecture, with cascade constant of the canonical Stuart-school form $\propto L_{\text{post}}^2/\rho_{\text{noise}}$ — tight at $2L_{\text{post}}^2/\rho_{\text{noise}}$ on conjugate-Gaussian and exponential-family-conjugate instances, with absolute multiplicative overhead from the structural kernel-tensorization step in general. Hosseini-Hsu-Taghvaei is the closest single-paper match for the amortized-inference framing; the structural-slot match holds for the entire strand.

Our distance from this strand: we apply this shared machinery to *belief-goal-coupled* inference under an architectural classification. None of these papers address goal-conditional bias under explicit architectural-coupling factorization. The relationship is shared-machinery, not theorem-containment — Strand 2 proves outputs (matching lower bounds, multi-metric extensions, sub-Lipschitz refinements, contraction rates, randomized forward models) outside Track 1’s averaged-over-goal reach.

1271 F.3 Strand 3: Fisher-Rao geometry for Bayesian sensitivity

Kurtek and Bharath 2015^[51] provide a Fisher-Rao framework for Bayesian sensitivity analysis under ε -contamination classes — the closest existing Fisher-Rao deployment to ours. Same geometric machinery; different target (sensitivity vs universal-constant bias bound). Lebanon 2004^[52] extends Čencov-Campbell uniqueness to conditional information geometry — load-bearing precursor for the conditional-distribution setting in which our bound lives. Čencov 1982^[18] is the original uniqueness result for the Fisher metric under invariance under sufficient statistics; Amari and Nagaoka 2000^[19] give the modern treatment of the KL-to-Fisher second-order expansion. McCulloch 1989^[53] and Ruggeri and Wasserman 1993^[54] develop earlier Fisher-information-flavored sensitivity analyses; both are foundational but produce neither a universal constant nor a goal-coupling bound.

1281 F.4 Strand 4: Bayesian brittleness (handle carefully)

Owhadi et al., Owhadi et al. 2015, 2015^[24,25] establish that Bayesian inference is brittle under finite information in continuous settings: arbitrarily close models on the same data can yield arbitrarily different posterior conclusions, and any prior + model can be perturbed slightly to achieve any desired posterior. Owhadi and Scovel 2017^[55] develops the qualitative-robustness companion analysis.

A reader might worry that the brittleness results contradict our Track 1 stability bound. They do not. Owhadi-Scovel-Sullivan’s no-go is for *arbitrary stability under finite information* in fixed metrics (TV, generalized moments) — they construct adversarial perturbations *outside* the well-posed regime. Our Theorem 4.1 (Track 1 instantiation) carries explicit (H1) + (H2’) hypotheses that exclude their brittleness regime: (H2’)’s post-update concentration is precisely the regularity that brittleness requires *failing*. The two no-gos constrain different things; our Theorem 4.4 is for *universal-constant absent the (PI) + (R) + (K) triple*, which is a third, distinct statement. The honest reading: three different no-go results, three different scope conditions, none implying or contradicting the others.

1294 F.5 Strand 5: Adjacent recent work

Sheng et al. 2025^[56] establish stability of mean-field variational inference under strongly log-concave targets — adjacent to Track 1’s transport-inequality machinery. Cattiaux and Guillin 2022^[57] develop functional inequalities (Poincaré, Cheeger, log-Sobolev) for perturbed measures with explicit constants — background machinery for Hypothesis 4.2. Ley et al. 2017^[58] give Wasserstein-distance bounds for nested densities via the Stein method. Cvetković et al. 2024^[59] address observation-operator choice for misspecified inverse problems. Zahm et al. 2022^[60] provide certified dimension reduction for nonlinear Bayesian inverse problems. Kleijn and van der Vaart 2006^[61] study misspecified Bayesian inference asymptotics; Shalizi 2009^[62] studies dynamics of Bayesian updating under dependent data and misspecified models. Seo et al. 2026^[63] derive W_2 -stability bounds for tangent-linearized Gaussian inference on smooth manifolds — adjacent for the manifold-Bayesian-inference framing.

In LLM-stability-flavored work: Xu 2025^[64] (*Policy Cliff*) analyzes RLHF stability for LLMs/LRMs through reward-policy maps; Biau and Boyer 2026^[65] develop statistical proxies for retrieval-augmented generation; Hyeon et al. 2026^[66] address action-sufficient goal representations in RL. None of these target the goal-coupling bias quantity (3) under an architectural-coupling factorization.

1309 F.6 Strand 6: Active inference and Markov-blanket apparatus

1310 The Markov-blanket apparatus from active inference^[67–69] supplies the conditional-independence
1311 machinery our architectural-separation classification invokes. We adopt the Pearl-blanket reading^[17]
1312 — the technical conditional-independence statement — and explicitly do not adopt the Friston-blanket
1313 metaphysical reading. The Goal/Update Coupling Class partition (Class 1 Separated / Class 2 Partial
1314 / Class 3 Coupled), with the explicit Coupled-class scope-exit, is the structural extension that the
1315 Pearl-blanket form invites but does not by itself produce.

1316 F.7 Relation to existing decompositions

1317 Zeng et al. 2026^[47]’s HalluGuard framework decomposes hallucination risk into *data-driven* and
1318 *reasoning-driven* components, linked respectively to training-time mismatches and inference-time
1319 instabilities. This is the closest-shape decomposition to our $\kappa_{\text{processing}} \times I$ factorization. The structural
1320 axes differ: HalluGuard’s axis is *temporal* (training-time vs inference-time errors); ours is *architectural-*
1321 *vs-informational* (where the goal can flow vs how much it could resolve). The two decompositions live
1322 at different layers and do not reduce to each other. A model can have low data-driven hallucination
1323 (clean training) and still exhibit large goal-coupling displacement (Coupled-class architecture in a
1324 high-ambiguity regime), and vice versa.

1325 Chlon et al. 2025^[46] frame transformers as *Bayesian in expectation but not in realization*, with
1326 permutation-induced order effects scaling as $O(\log n)$. This is structurally distinct from our framing
1327 — we treat the architecture as performing a single goal-conditioned belief update on a fixed input,
1328 not as averaging over orderings — but the two are compatible: their Quantified Martingale Violation
1329 bounds the deviation from Bayesian-in-realization behavior, and our bound captures one specific
1330 channel (goal-coupling) of how non-Bayesian-in-realization a coupled architecture can be.

1331 F.8 Failed routes — pointer

1332 Two derivation routes failed at structural levels and are documented in Appendix A: (F1) Cramér-Rao
1333 inversion fails because Cramér-Rao bounds estimator variance from below, and inverting it for an upper
1334 bound on side-information-induced displacement requires a fixed-estimator-class assumption not
1335 available in coupled-architecture settings. (F2) Rate-distortion inversion fails because rate-distortion
1336 theory describes the minimum bits-per-symbol required to *represent* a source within distortion D ,
1337 not the maximum *displacement induced* by injecting side-information into an update. The pattern:
1338 information-theoretic *lower-bound* machinery for estimator or encoder performance does not invert to
1339 upper bounds on side-information-induced displacement.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Section 1 state the four contributions: (i) the architectural Goal/Update Coupling Class partition, (ii) the umbrella upper bound on goal-conditional displacement of the post-update model state with two named tracks supplying the constant, (iii) the no-go theorem on Euclidean chart norms with the locally-tight $\sqrt{2}$ and globally-valid $C = 2$ Fisher-Rao constants forced under (PI)+(R)+(K) and (PI) respectively, and (iv) the architectural-corollary factorization $\kappa_{\text{processing}} \times I(G; \Omega \mid e, M)$ under (H_κ) . Each is developed formally in Sections 3–5 and proved in full in Appendix E.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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